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Evaluating Energy Efficiency and Optimal Positioning of Industrial Robots in Sustainable Manufacturing

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Abstract: Optimizing the energy efficiency of robotic workstations is a key aspect of industrial automation. This study focuses on the analysis of the relationship between the position of the robot base and its energy consumption and time aspects. A number of 6-axis robots, including the ABB IRB 120 robot, were investigated in this research by combining measurements and simulations using the energy consumption measurement module in the ABB RobotStudio 2024.1.1 environment. The objective of this study was to develop an energy consumption model that can identify the optimal robot positions to minimize energy costs and time losses. The results suggest that the strategic positioning of the robot has a significant impact on its performance and efficiency. These results demonstrate that the ideal working distance of the robots is approximately 50% of its maximum range, and displacements along the X and Z axes affect the energy and time consumption. These findings suggest the existence of a trade-off between time and energy efficiency, providing a basis for further research into the optimization of robotic systems. Thus, this work offers new perspectives for the design of efficient robotic workstations for cross-sensory applications.

Keywords: energy consumption; industrial robot; simulation; sustainability; workplace layout

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1. Introduction

The current manufacturing trends are robotization, automation, and digitalization [1,2]. Robotic applications are widespread worldwide, and 593,000 robots were deployed in 2023. In 2024, the number of deployed robots is projected to be 622,000, representing an increase of 7% from 2023. In the following years, the assumption is set that the deployment of robotic applications will increase by 7% annually [3]. In this context, advances in smart manufacturing and Industry 4.0 strategies have made industrial robots a key part of automated and intelligent production.

As the adoption of robotic applications in manufacturing continues to grow, which is driven by technological advances and evolving market needs, the industry faces an urgent challenge: to dramatically reduce the time required to design and deploy these systems, while ensuring that increasingly sophisticated, multi-sensor robotic systems operate flawlessly and efficiently from the outset.

The proper planning of investments in robotics is important because robotic applications require large investments. Therefore, it is important to manage design and planning processes in order to improve the efficiency and economic flexibility of the solution.

In the field of robotic workstation design, layout refers to the strategic arrangement of components within a manufacturing system, encompassing the robot's base position, fixtures, and the workpiece that the robot processes or manipulates. An optimized layout enhances the workstation efficiency by minimizing the travel distance and time required for the robot's movements, thus reducing both the energy consumption and the operational time.

Research and practical applications indicate that an effective layout is essential for streamlining production processes. A well-designed layout not only optimizes energy use but also positively affects the system's operational dynamics, enabling shorter cycle times and improving ergonomics. This setup promotes safety and productivity, making it a crucial factor in industrial robotics.

In the planning and simulation of robotic workstations, layout is a key consideration for achieving optimal efficiency. Careful positioning of the robot relative to the workpiece and fixtures allows for configurations that lower energy demand while maximizing productivity, making layout a foundational aspect of designing energy-efficient, high-performance robotic workstations.

Poorly designed robotic workplace layouts can cause damage, can reduce workplace productivity, and may not be worthwhile. Consequently, corrections and changes occur in the design. It makes the design itself more expensive. Conversely, a well-designed layout can confirm the correctness of investments in robotics, ensure higher productivity and quality, and protect people from strenuous activities.

Despite the high demand for robotic applications, humanity is on the threshold of an energy and ecological crisis [4], and the consumption of electricity itself must be addressed because robotic devices consume a significant portion of energy, which is reflected in the price of the final product [5]. Any solution that can help save energy will be significant in the future [6].

A well-designed robotic workplace can significantly contribute to both energy savings and required productivity [7,8]. Due to the complexity of the robotic workplace design process, there is no possibility of creating workwise scenarios using automated functions in simulation software [9]. Companies involved in the design of such workplaces only use already developed functions that can semi-automatically generate the position of the robot relative to the periphery or generate a path based on the shape of a digital model [9,10].

Furthermore, the design process was conducted by using the methodology and simulation software used in practice [11,12]. There are two main principles to be followed in the design process: maintaining the predetermined work tact time [13] and the reachability of the work points by the robot [14,15]. These two principles can already be observed in the first design phase when deploying equipment in a robotic workplace.

A suitable layout [16] can reduce the working time and energy consumption of the workplace. Not only the layout of components but also the appropriate software and hardware setup of industrial robots can significantly reduce the energy consumption of robotic workplaces [17–19].

Industrial companies contribute 36% of global energy consumption and 24% of CO_2 emissions [20]. Companies are trying to reduce greenhouse gas emissions [21,22] and achieve an environmentally sustainable future [23,24]. In addition to time targets, energy consumption targets are important in production-planning decisions [25,26].

According to Fysikopoulos et al. [27], 20–40% of unnecessary energy consumption can still be found in industrial firms. This is the reason why the energy performance of manufacturing systems is a major research area and problem for many manufacturing companies [28].

The automotive industry places a strong emphasis on product energy efficiency. And, particularly in the automotive industry, the energy required for production is becoming increasingly important [29,30]. This phenomenon can be clearly demonstrated by the fact that the average energy consumption to produce a saloon car is 3.5 MWh, which is sufficient for a range of up to 27,000 km for an advanced electric car. The entire process chain needs to be considered to identify and ultimately realize the energy-saving potential of vehicle production. Not only do press and paint shops appear to be particularly energy-intensive, but body shops are also identified as requiring large amounts of energy [31].

The high energy consumption of the body shop was mainly due to the energy consumption of the cross-machine robots. Generally, the power consumption of industrial robots in the automotive industry is approximately 8%. For example, a typical angular

industrial robot with a load of 200 kg requires power of 0.5 kW in standby re-mode and up to 20 kW in peak mode. This is highly dependent on the specific robot type, application, tool, workpiece, motion trajectory, usage strategy, and many other factors. They together create a complex set of influencing variables [32]. The appropriate design and operation of automation systems and industrial robots present a good opportunity for reducing energy consumption in industry, as well as energy optimization of operations and replacing more efficient systems [33].

2. Background

According to available literature and research, energy optimization can be improved by appropriate software or hardware settings or by optimizing the workplace layout. Energy optimization can be addressed in various ways, including mechanical and dynamic optimization or by modifying the mechanical parts of the robot [34–37]. Research is limited to available industrial robots without the possibility of modifying their structure and mechanics.

2.1. Theoretical Framework for the Energy Consumption of Industrial Robots

This research relies on a theoretical framework based on the fundamentals of mechanics, kinematics, and the dynamics of industrial robots. In this case, it is an industrial robot with articulated kinematics in a serial configuration [38,39].

One of the key factors influencing the energy consumption of the robot is the moment of inertia (I), which measures the robot's resistance to changes in its rotational motion. In the case of articulated robotic arms, the moment of inertia depends on the weight distribution of the individual segments and their distance from the rotation axis [40]. Mathematically, the moment of inertia is expressed as follows:

$$I = \sum_{i} m_i * r_i^2 \quad (1)$$

where m_i is the mass of segment i, and r_i is the distance of segment i from the rotation axis [41].

In practical terms, a higher moment of inertia indicates higher energy requirements when accelerating and decelerating the arm. Therefore, the optimal TCP position configuration minimizes the moment of inertia. The principle of minimizing the moment of inertia is universally valid for most articulated robots and provides a basic theoretical framework for understanding the energy requirements of different TCP positions [41].

For vertical robot motions, gravity plays a significant role as it requires additional torque to overcome gravity. The torque τ required to move upwards against gravity can be calculated as follows:

$$\tau = m * g * h \quad (2)$$

where m is the total mass of the robot segment, g is gravitational acceleration, and h is the height of the motion.

For upward movements, the energy requirement is higher because the robot has to work against gravity. In contrast, for downward movements, the robot can use energy recuperation, which allows part of the potential and kinetic energy to be converted back into usable energy. This principle of energy recovery can significantly reduce energy consumption during controlled downward motion.

These principles demonstrate that when designing robot trajectories, vertical motions should be minimized to reduce energy consumption [42].

Some robots have balancing mechanisms (e.g., springs or counterweights) that compensate for gravity loads during vertical movements, thereby reducing the need for additional torque and energy requirements. These mechanisms are designed to reduce the moment of inertia during upward or downward motion and are typical of robotic arms used for handling heavier loads on vertical axes [43,44].

Robots equipped with these balancing mechanisms can perform vertical movements more efficiently. Conversely, robots without such systems exhibit higher energy consumption during lifting because they do not have additional support for weight balancing. The experimental results suggest that robots with similar balancing mechanisms achieve similar energy consumption results, allowing for the generalization of these findings to other robots with similar balancing systems [45].

The optimization of the robot's position in terms of energy efficiency is based on the goal of setting the optimal TCP distance in the robot's workspace. The optimal TCP position minimizes the moment of inertia and reduces the torque requirements, thus consuming less energy during movements and accelerations. This approach leads to an overall optimization of the system's energy consumption [46]. Mathematically, the position optimization for minimum energy consumption can be formulated as an optimization problem:

$$min_{\theta}(\tau)\ for\ a\ given\ trajectory, where\ \tau = I * \alpha \quad (3)$$

where θ represents the angle of the robot segments, τ is the torque, and I is the moment of inertia.

The theoretical principles and assumptions above provide a deeper framework for understanding the impact of design and kinematic parameters on the energy efficiency of the robot. However, in this approach, this research focuses on optimizing the position of the robot's basic coordinate system with respect to the TCP and the trajectory being generated. This method allows people to understand why energy requirements vary depending on the relative position of the robot and the TCP and why some of the configurations may be more advantageous in terms of energy consumption.

The experimental validation of these results could allow the application to different types of robots with similar kinematics and generalization of the optimization rules to a wider range of robotic systems in industrial environments. The optimization is based on a balance between the distance of the TCP from the axis of rotation and the minimization of the post-torque required, thus reducing the overall energy demand of the motions.

Testing the optimal placement of the robot coordinate system relative to the TCP and trajectory often yields favorable results, with the most accurate configuration for a particular application determined experimentally.

2.2. Software Settings to Reduce Energy Consumption

The software optimization is based on the idea that the energy consumption in industrial production facilities results primarily from the control and operation of electrical drives in automated production processes [47]. To maximize production, robots and machines are often operated dynamically. This causes high-energy losses during high-speed movements and excess energy loss during deceleration [48]. In addition, many axis movements are followed by periods of inactivity with a loss of productivity.

One of the ways to save energy is when the robot is in motion. For this reason, several methods have been proposed to plan energy-optimal trajectories [49,50].

However, industrial robots can save energy even when they are not moving. Unlike modern electronic devices, there is a lack of efficient and programmable options for switching off robots in idle phases [51].

In 2011, a patent was published describing an energy-saving function for industrial robots when idle [52]. Some robots, such as KUKA robots, support the PROFIenergy profile, which allows the robot to control energy consumption by sending commands over the PROFINET network. The latter mode allows power-saving modes for robots such as bus shutdown or hibernation.

Several experimental facilities use PROFIenergy, such as the Mercedes-Benz plant in Sindelfingen, where various experiments have been conducted [53].

Meike and Ribicks [32] presented the following sample examples showing potential annual energy savings by the type of stabilizing frame:

- Downtime without production. Reducing the standard consumption of 275 W only on weekends provides an average of 700 kWh per year per robot or 8.7% of its total consumption.
- Shutdown during periods of inactivity lasting longer than 10 min. The PROFIenergy protocol developed at SIEMENS AG, which is based on ProfiNET, now enables controlled shutdown, leaving only the power supply required for the machine's network communication. Using it for shorter shutdowns, for example, from 10 min to 3 h, would yield an additional 400 kWh of savings or 5% of the total annual consumption.
- Low power standby. The PROFIenergy protocol also allows the machine to enter a known standby state when the pause time is below the minimum restart time of the device. In industrial robots, this shuts down the robot's servo motor amplifiers, which supply an additional 50 kWh or 0.6% of the electricity per year.

The problem between the last two points and downtime on weekends is that short-term breaks and downtime are often not known in advance, which reduces the usable potential. The software settings when the robot is in motion can be divided into two categories: trajectory optimization and operation scheduling [33].

2.2.1. Trajectory Optimization

The trajectory optimization can be described in more detail in the split groups. The first group focused on the spatial adjustment of the trajectory, i.e., the way the robotic arm performs its movements. The second group focuses on the velocity and acceleration settings [54,55].

Currently, most research on the energy conservation of industrial robots is focused on improving their trajectory and path planning. The speed and acceleration factors also significantly affected the power consumption. For example, slow motion of industrial robots consumes more energy than fast motion [51].

The trajectory is one of the most significant aspects that affect the energy consumption of industrial robots. During the execution of the work of industrial robots, factors such as direction and speed of motion, acceleration, and inertia affect the power consumption [56].

A typical industrial robot has a configuration with a 6-axis as the articulated robot [57]. Each joint is rotated by an electric motor about its axis. At each time instant, an angular position is associated with each joint, and the set of positions of all joints provides a complete picture of the robot's position at that instant, which is referred to hereafter as the robot's position. A straight-sided robot repeatedly performs several operations, each following a trajectory of motion. The trajectory describes the synchronous movement of all robot joints through time [58].

Motion optimization can be used to reduce various parameters, such as the time, amount of vibration, and energy consumption of mechatronic and robotic systems [59].

The track and movement profiles were modified to increase energy efficiency. An important aspect of this approach is its application to existing systems without requiring modification or redesigning hardware. With respect to trajectory, two main distinct groups can be defined: point-to-point trajectory (PTP) and multi-point trajectory (MP) [60]. The applicability of these methods to different robotic and mechatronic systems depends on the complexity of the kinematics and the level of detail of the considered dynamical model [33].

- Point-To-Point (PTP) trajectory optimization. PTP trajectory optimization methods can be divided into inverse and forward kinematic approaches. In the inverse kinematics approach, the mechatronic and robotic systems are developed through a mathematical model that is simplified in all-absolute to reduce the computational complexity. The trajectory, which is usually expressed as a polynomial function, is then synthesized with respect to the energy content and minimized during optimization. Conversely, forward kinematics defines the mechatronic and robotic systems in a more complete way without neglecting secular losses (e.g., motor losses, motor drive losses, and auxiliary device losses) and considering the non-ideal behavior of the losses because non-ideal losses depend on the motion parameters. The model is then used to test

different motion planning strategies (i.e., different predefined motion profiles) and to assess the impact of different parameters on energy consumption through sensitivity analysis [33].

- Multi-Point (MP) trajectory optimization. In MP trajectory optimization, the energy consumption of the mechatronic system is achieved by optimizing both trajectory and timing. Both planar and spatial kinematics are used to achieve planar motion. In general, the parameters are obtained using a data sheet or CAD, or they are estimated based on some assumptions. Rarely, the dynamic parameters are obtained through an identification procedure in which the robot performs a specific trajectory, and the torque, velocity, and acceleration data are collected directly or indirectly before the dynamic parameters are estimated [33].

The literature review demonstrates that by optimizing the paths it is possible to reduce energy consumption. Roßmann et al. [59] generated a method for moving along a path using AI. Their simulation showed that the energy consumption decreased from 24.5 to 3.5 J. Rassőlkin et al. [61] found that the energy consumed by an industrial robot is higher when moving in the vertical direction than when moving in the horizontal direction.

For this study, an ABB IRB 1600 robot was used, operating at a specified movement speed of 500 mm/s with a tool weight of 2.5 kg. During horizontal motion, the robot exhibited a power consumption of 11.4 Wh, whereas during vertical motion, it exhibited a slightly lower power consumption of 10.4 Wh. The trajectory was carefully designed to engage all six axes of the robotic arm throughout the movement to ensure a comprehensive assessment of the robot's energy requirements across different orientations.

This research illustrates two motion patterns of an ABB IRB 1600 robot: horizontal and vertical. In the horizontal motion setup, the robot's end effector follows a figure-eight trajectory along a horizontal line with key points P1, P2, and P3. The coordinate system at the robot's base is oriented with the X-axis forward, the Y-axis to the side, and the Z-axis upward. This configuration allows the robot arm to oscillate horizontally by using all six axes to achieve smooth motion across the horizontal plane.

In the vertical motion setup, the robot's end effector follows a similar figure-eight pattern but along a vertical line, with P1 positioned in the middle, P2 above, and P3 below. The coordinate system remains consistent with the horizontal setup. This configuration demonstrates the robot's ability to perform oscillating vertical motion, which involves different dynamics and energy requirements than horizontal movement.

This comparison of horizontal and vertical motions highlights the impact of trajectory orientation on the robot's power consumption, providing insights into optimizing the energy efficiency based on the movement direction and load distribution. The movement from a straight-ahead position to a horizontal position is associated with reduced energy consumption because the robot arm is accelerated by gravity. Movement in the horizontal–horizontal plane requires nearly constant power at low speeds ($v \approx 20\%$) regardless of acceleration.

The significant point is the beginning of the movement from horizontal to upright with acceleration against gravity, where high power is initially required, which slowly decreases. The phase of rotation back along the upright position is comparable to motion in the horizontal plane with much less inertia [51].

2.2.2. Effects of Speed and Acceleration

Slower movements require much more energy than fast ones. This is because the execution time of slow movements is more than 10 times longer than that of fast post-movements, whereas fast movements have only short peaks of high energy consumption [51].

Rassõlkin et al. [61] moved the manipulator 50 times along a pre-programmed path at different speeds during their experiments. The speed was increased in 100 mm/s increments to a maximum.

The measurement conditions included a horizontal plane of motion and a tool weight of 2.5 kg. Under these conditions, the results indicated that the optimal operating speed

for minimizing energy consumption was 600 mm/s, at which the robot demonstrated the lowest energy consumption. This speed highlights an efficient balance between speed and energy demand in horizontal motion with the specified load, providing a basis for determining ideal operational speeds to enhance the robot's energy efficiency in similar applications.

The results also revealed a clear relationship between the speed of the ABB IRB 1600 robot and its energy consumption. At lower speeds, particularly around 100 mm/s, the energy consumption was significantly higher, exceeding 20 Wh. As the speed increased, the energy consumption decreased steadily, with notable reductions observed up to approximately 300 mm/s. Beyond this point, the energy consumption leveled off and remained relatively stable, showing only minor fluctuations with further increases in speed. Even at maximum speed, energy consumption did not return to the high levels observed at the lowest speeds.

This trend suggests that the robot operates more efficiently at moderate to higher speeds, with lower energy requirements as the speed increases, indicating an optimal range for energy efficiency. These findings support the identification of specific speed ranges that minimize energy consumption, offering valuable insights for optimizing robotic operations in industrial settings.

The research by [62] demonstrates that energy consumption in robotic systems can be minimized by carefully selecting specific combinations of speed and acceleration limits. Their model shows that certain ranges of these parameters lead to lower energy requirements while maintaining efficient movement times. As speed and acceleration increase, energy consumption generally rises; however, there is an optimal range where energy demand is minimized without compromising performance.

These findings highlight the potential for energy savings in robotic operations through strategic adjustment of speed and acceleration, allowing for effective balancing of energy efficiency with operational effectiveness. The proposed approach provides a practical framework for optimizing energy use in industrial applications by selecting appropriate motion.

Bukata et al. [53] have developed a mathematical model considering operational parameters to save energy for a robotic cell by optimizing robot speeds and positions in a robotic cell. Their work reduced energy consumption by approximately 20%.

Although reducing the movement speed may minimize the robot's energy consumption, it increases the time required to perform the operation, which is contrary to the stated goals. Energy savings in terms of speed reduction are not always conceivable, especially regarding mass production, where cycle time is crucial [63].

2.2.3. Operations Planning

Operation scheduling is another approach related to motion scheduling modification that saves energy by reducing the energy consumption of a single robotic system or cell [64].

In a robotic cell, the robot works synchronously with other robots or mechatronic systems, and the entire work cycle must be balanced to avoid downtime. In addition, in some operations, the robots must share the same work area. Thus, the robot waits for the previous robot operation to complete before performing its task to avoid collisions.

The scheduling approach focuses on optimizing the operations of the robotic cell, and two main methods can be found: time scaling and sequential scheduling [33].

- Time scaling. This method is because, in most cases, the industrial robot is programmed to execute the operation as quickly as possible (i.e., with the shortest cycle time without taking energy consumption into account) and with a consistent idle time or downtime between two operations until the next operation is started. This approach implements the idea that a reasonable reduction in energy consumption can be achieved by maximizing the deceleration rate (i.e., reducing the mean and peak acceleration). The consequent increase in the operating time is compensated by increasing the slow movements and reducing the idle time; thus, the duty cycle time is not affected [32].

- Sequential planning. The goal is to find a more energy-efficient sequence of operations by maintaining a fixed cycle time. It also aims at reducing the acceleration load in the production cell by processing and optimizing a given work plan in cycle time and by guaranteeing the avoidance of collisions and blockages. The idle time between different operations was then used to reduce the speeds and acceleration to reduce the energy consumption [65].

Riazi et al. [58] demonstrated that by minimizing the speedup and optimizing the scheduling of operations, energy consumption can be significantly reduced. The optimization algorithm was implemented and validated for a multi-robot system, including the ability to perform new operations and their trajectories in real robots. The results demonstrate the possibility of reducing energy consumption by up to 45%. The energy is saved by changing the trajectory [66] of the robot, which slows down the movements to synchronize better.

2.3. Hardware Settings

Hardware optimization includes the implementation of new types of drive systems and energy recovery and distribution strategies. This enables a new approach to reduce energy consumption and thus reduce environmental impact. By selecting an appropriate automation system for the application, it is possible to reduce energy consumption while maintaining productivity. The hardware setup for energy efficiency can be further divided into the following three subcategories [33]:

- Robot-type. By selecting the appropriate automation system for a given application, it is possible to achieve a reduction in energy consumption while maintaining productivity [33]. Industrial robots use only two types of kinematic pairs (links) in their kinematic chains: rotary (R) and sliding (T). Depending on which axis of the coordinate system the motion is relative to, an index is added to the base marker—Rx, Ry, Rz, Tx, Ty, and Tz. To determine the position of a body in space, 6 independent coordinates are needed [67].
- Replacing hardware components. Another strategy under consideration is the lighter design of components to reduce the moving masses, which means reducing the weight and inertia of the arms. In this way, the torque can be reduced, thereby reducing the energy consumption [33]. This can be achieved by using suitable materials such as carbon fiber or by optimal design of a serial robotic manipulator [68]. In the latter case, the goal is to find the optimal diameter of the arms and the optimal dimensions of the relief holes in order to minimize the weight without adversely affecting the structural–mechanical behavior [69].
- Additional hardware components. One of the latest techniques adopted to reduce the energy consumption of mechatronic and robotic systems is the addition of energy storage and regeneration devices. The main idea behind these devices is to collect energy during the braking phase and return it to the system when needed [70].

Based on an analysis of the current state-of-the-art technology in the field of energy and the time reduction of industrial robots, the importance and interest in the problem addressed in this article is evident. In the introduction of this chapter, the methods of energy intensity reduction are divided into three categories. The first and most scientifically investigated concerns the software setup of industrial robots. This area is primarily concerned with optimizing robot trajectories and adjusting their velocity and acceleration parameters. The second category deals with hardware or design modifications to industrial robots. Here, the goal is to reduce energy consumption by making additional modifications to the equipment or by adding an external device to share energy between the devices. The last and least common category in the energy intensity area is the layout of the workplace. Most research on robotic workplace layouts addresses the layout problem with respect to work cycle time reduction. Only a minimum addresses the dependence of the layout on power consumption.

For research with a focus similar to this article, one can highlight the research performed by Rassõlkin et al. [61].They partly dealt with the change in the working height along the Z-axis and its influence on electrical consumption. However, it should be noted that their meta-analysis was carried out on only one industrial robot, which changed its working height along the Z-axis when performing a nav-designed trajectory in the shape of a figure of eight. Thus, it can be concluded that the measurement of energy consumption was not only performed at the chosen level according to the author's state.

2.4. Motivation

In recent decades, the industry has focused on the deployment of robotic systems to increase the speed and volume of production, thereby improving the efficiency and competitiveness of businesses. However, with this boom in automation comes challenges in the form of higher energy demands, which place a significant strain on environmental resources and increase the cost of production. Industrial robots, equipped with sophisticated sensor systems, cameras, and specialized peripherals, are themselves energy-intensive, directly affecting not only the economic efficiency of the production process but also the environmental footprint of the enterprises concerned. In this context, it is important to optimize the placement of the robot to minimize energy consumption during its working cycle. Reducing the energy requirements of robotic workplaces is, therefore, a pressing issue not only from a cost perspective but also with respect to sustainability and reducing the carbon footprint of production. From this perspective, workplaces that not only meet production requirements but also contribute to long-term environmental sustainability must be designed.

In the HUB4.0 lab, this issue has been addressed for a long time, exploring ways to reduce the energy intensity of robotic workplaces. The goal of this article is to find solutions that not only reduce costs but also contribute to environmental protection and the achievement of carbon neutrality goals in industry. Therefore, the motivation of this research is to find the optimal position of the robot with respect to the planned trajectory that would ensure the lowest possible energy consumption without sacrificing the time efficiency of the work process.

The main question of this research is how the position of the base of a 6-axis articulated robot affects its energy consumption and time requirements. Another time-part question is how this knowledge can be used to optimize workplaces for energy efficiency.

The first objective of the research is to investigate in detail the relationship between the position of the robot's base and its overall energy and time requirements. This objective includes an in-depth analysis of how different robot base positions within the working range affect energy efficiency, not only in terms of energy consumption but also in terms of the overall time consumption of individual robot movements.

The second objective is to determine the optimal position of the robot base with respect to the all-absolutely defined trajectory. The task is to determine the position of the base that allows efficient use of energy during transfers and execution of work tasks without negatively impacting the time efficiency of the process. Thus, it is a matter of determining a configuration that combines minimization of energy consumption with time efficiency.

The first supporting research question is what the optimal position of the robot's terminal point (TCP) within its working range that would minimize energy consumption. The goal is to experimentally verify whether theoretical factors such as moment of inertia and gravitational load affect the energy efficiency of the TCP and to investigate whether these factors can be used to accurately determine the optimal position for different applications.

The second supporting question focuses on the extent to which the TCP optimization results for energy efficiency can be applied to different types of 6-axis robots with similar kinematic structures. This section focuses on the experimental validation of whether the findings valid for one robot type can be applied to other robots with similar mechanical structures, which allows us to formulate general conclusions for optimizing energy efficiency across industries.

Given the HUB4.0 lab's long-standing interest in optimizing the energy efficiency of robotic workstations, the present research is expected to yield deeper insights into the impact of the robot's base position on its energy and time requirements. The strategic positioning of the robot's base is expected to have a major impact on the efficiency of its movements, whereby optimal positions can significantly reduce energy consumption while improving the time performance of individual operations.

Specifically, it is expected that optimizing the position of the base along the X and Z axes will enable a balance to be struck between time and energy efficiency, which may be beneficial for companies considering robotizing workplaces to reduce costs and the ecological footprint. The research should also confirm that an appropriate workplace layout can achieve lower energy intensity at an early stage of design, opening the way for greener production and meeting carbon neutrality targets.

An important part of the research will be to experimentally validate whether the TCP optimization results can be applied to different types of 6-axis robots with similar kinematic structures. If these results are confirmed to be generalizable, it will be possible to formulate general conclusions for optimizing energy efficiency in industrial applications. These expected results could be of significant relevance to the automotive and other industries where optimizing the energy consumption of robotic workstations can significantly reduce overall operating costs while contributing to environmental sustainability.

3. Materials and Methods

This study is based on a theoretical framework and previous research results in the field of energy consumption optimization of industrial robots. The results and theoretical insights led us to formulate the main research question and objectives and to choose a qualitative research approach that combines simulation and experimental models. The methodology consists of several steps, which are detailed to ensure the credibility of the results and to form a coherent procedure to investigate the effective position of the robot in terms of its energy efficiency. The Materials and Methods section is elaborated in detail to make the research as reproducible and verifiable as possible for future studies.

In the first phase of the research, the focus is on verifying the accuracy of the simulated values by comparing them with the actual measured values of the ABB IRB 120 industrial robot [71]. This step is crucial to confirm or refute the reliability of the chosen method for determining the optimal robot position in terms of energy consumption. For this purpose, the energy consumption of the real robot was measured and compared with the simulation results in the ABB RobotStudio 2024.1.1 environment. Validation of the confidence of the values allowed us to assess whether the simulations and the actual measurements show consistent results, ensuring that the values from the simulation accurately reflect the real energy consumption.

After successfully verifying the validity of the simulations, experiments continued with the simulation models, focusing on a broader sample of 6-axis industrial robots with articulated structures, similar to the ABB IRB 120. The robots were selected to represent a variety of configurations to allow us to investigate the effect of base position on energy consumption and time requirements. To ensure consistency of measurements, we designed a parametric trajectory that was consistent across all models and allowed for standardized measurements, applicable across robots with identical or similar designs.

Based on the parametric trajectory, energy and time consumption measurements were performed for each robot model in the ABB RobotStudio 2024.1.1 [72] and Siemens Tecnomatix Process Simulate 2402 [73] simulation environments. The robots were operated without tools in these simulations, which allowed the authors to obtain baseline energy consumption data under ideal, controlled conditions. Such conditions eliminate the influence of other factors, such as load or attached tools, and allow the authors to focus solely on the effect of base position on energy efficiency.

After collecting the energy and time consumption data for each robot model, the authors performed an analysis to identify the optimal robot base position that minimizes

the energy consumption without negatively impacting the time efficiency. The results provide valuable insights into how changes in the base position affect the energy consumption of robots, enabling the design of workplaces that are energy efficient and promote environmental sustainability. The approach and energy consumption model have been designed and described in detail so that they can be reproduced and validated in further studies, providing practical solutions for various industries such as automotive.

3.1. Limitations

This study has two major limitations that provide scope for future research. First, the analysis was conducted exclusively on ABB's 6-axis articulated robots, with the physical sample of the ABB IRB 120 supplemented only by simulated models of other robots of similar design.

This limits the generalizability of the results to robots manufactured by other manufacturers or to robots with different architectures. Although the conclusions are relevant to the samples studied, further research could include robots from different manufacturers or with different designs to broaden the applicability of the findings.

Second, the measurements were performed in a simulated environment without any loads or instruments to represent the ideal conditions. This approach allowed us to analyze the baseline energy consumption but did not take into account the energy demands of real operating sub-conditions with different loads or tools. Future studies involving physical experiments with real loads may yield even more accurate data and extend our conclusions to more practical scenarios.

However, these limitations do not diminish the importance of our findings on optimizing robot positioning in terms of energy intensity and provide a basis for future research that can further develop these aspects.

3.2. Choosing a Robot Model

For this research, an ABB IRB 120 industrial robot, which is available in the HUB4.0 laboratory, was chosen as the object of investigation. In order to consider the power consumption and time data from the simulation as relevant, it was necessary to perform a comparison between the data from the simulation and those from the real robot.

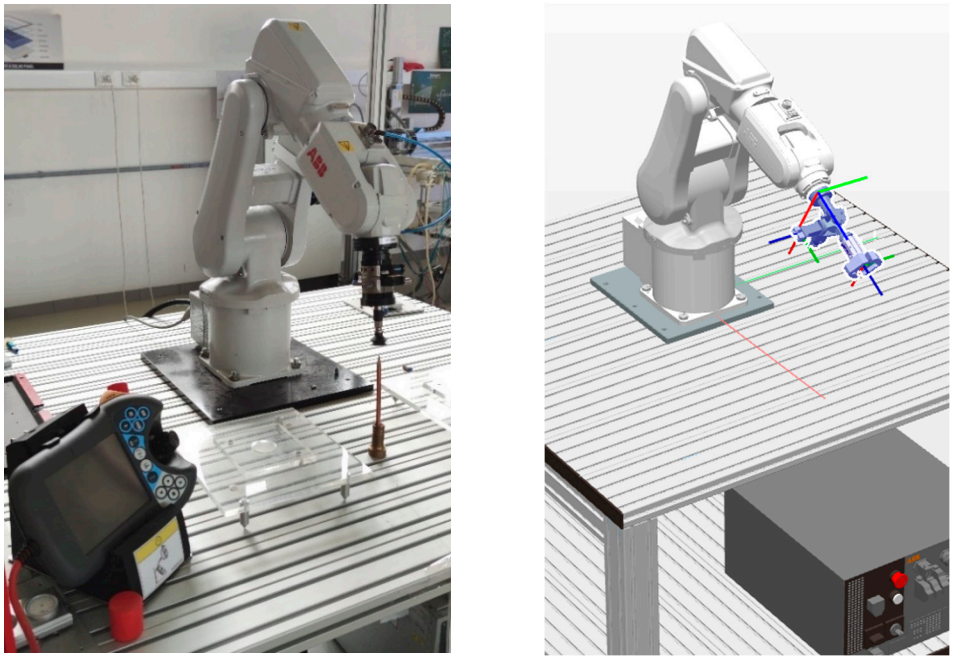
For this purpose, an ABB IRB 120 industrial robot (Figure 1a) was used in the HUB 4.0 laboratory at our faculty. A simulation model was also created for this (Figure 1b). The robot has an arm length of 0.6 m and a payload of up to 3 kg.

3.3. Robot Model—Kinematics D-H

The mathematical model of the robot used for energy consumption analysis is, in most cases, reused in numerical simulations as a virtual model. The implementation of an identical model for control design and simulation tasks leads to idealized results that may not accurately reflect realistic operating conditions.

The IRB 120 robot, which is one of the industrial robots with 6 degrees of freedom (6-DOF) manufactured by ABB, was designed with six rotary joints. The robot configuration (Figure 2a), including the attached frames, is shown in Figure 2b.

An industrial robot has specified types and ranges of motion for each joint of the ABB IRB 120 robot for a range of movement and velocity [74]. The ABB IRB 120 robot is equipped with six axes, each with specific motion types, ranges, and velocities that enable precise operation. Axis 1 performs rotational motion with a range from -165° to 165° and a speed of $250^\circ/s$. Axis 2 and Axis 3, which control the arm movements, have ranges of -110° to 110° and -110° to 70°, respectively, both operating at $250^\circ/s$. Axis 4, responsible for wrist rotation, moves within a range of -160° to 160° at $320^\circ/s$. Axis 5 provides a bending motion, with a range from -120° to 120° and a speed of $320^\circ/s$. Finally, Axis 6 enables turning with an extensive range of -400° to 400° and a speed of $420^\circ/s$. These parameters collectively contribute to the robot's flexibility and high precision in various industrial applications.



(**a**) (**b**)

Figure 1. Comparison of the ABB IRB 120 industrial robot setup: (**a**) the physical robotic workstation, featuring the actual robot mounted on an aluminum profile table with its end-effector and tooling, alongside the teach pendant used for programming and control; (**b**) the corresponding simulation model of the workstation in ABB RobotStudio 2024.1.1, demonstrating the robot's configuration, tooling, and workspace for virtual task planning and analysis. This comparison illustrates the alignment between the physical setup and its simulation environment for research purposes. The colored axes in the image represent the TCP (Tool Center Point) coordinate system, where the blue axis indicates the Z-direction, the green axis represents the Y-direction, and the red axis represents the X-direction.

For this purpose, the forward kinematics is described as the Denavit–Hartenberg (D-H) representation [75].

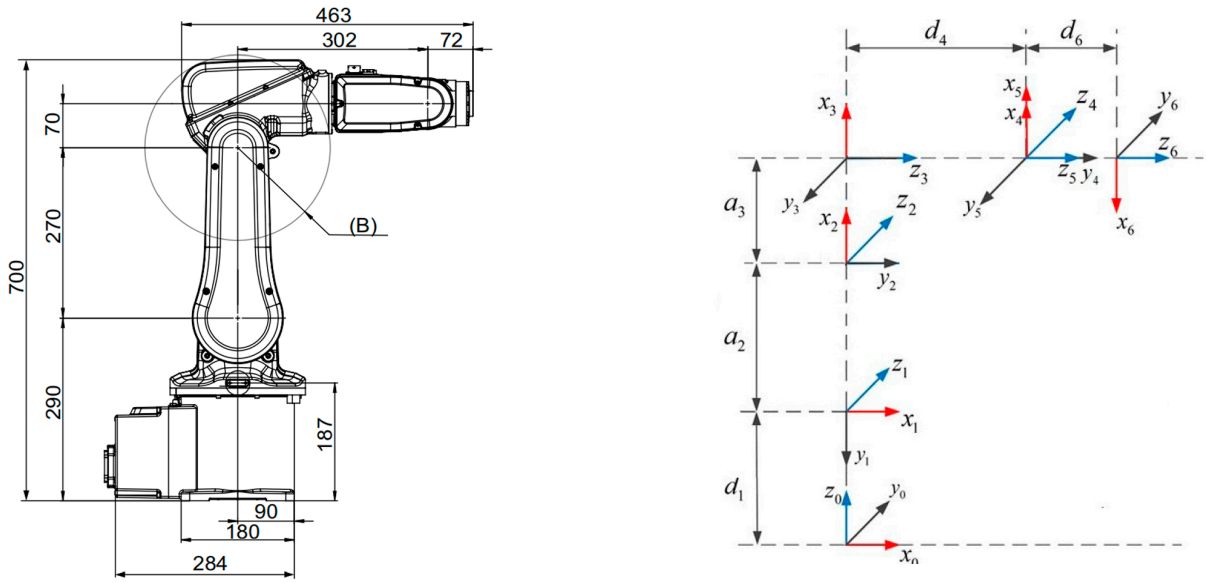
Kinematic analysis of robotic manipulators can be divided into two main areas: direct and inverse kinematics [76–79].

Direct kinematics analyzes the positions and orientations of links based on a specified set of angular joint positions ($\theta_1, \theta_2, \ldots, \theta_n$).

This concept is used as the basis for the geometric representation of the system and is considered as the starting point for the solution of the inverse kinematics of the manipulator, where the necessary angular positions of the joints are determined to achieve the desired position and orientation of the effector. The defined parametric trajectory can be solved via inverse kinematics in a simulation model.

Although the kinematic representation of robotic manipulators has been studied by several approaches, the formulation used in this analysis is that introduced by Denavit and Hartenberg.

Once the coordinate system is established, the robot structure can be represented by four D-H parameters associated with each link: joint angle (θ_i), link length (d_i), displacement distance (a_i), and rotation angle (α_i). Table 1 presents the D-H parameters associated with each robot link (Table 1).



(**a**) (**b**)

Figure 2. Geometric and kinematic representation of the ABB IRB 120 industrial robot: (**a**) detailed schematic showing the robot's geometric parameters, including key dimensions and distances between pivot points, useful for understanding the mechanical structure and workspace; (**b**) visualization of the robot's link coordinate system, illustrating the Denavit–Hartenberg (D-H) parameters and the orientation of individual joint frames, which form the basis for kinematic and dynamic analysis.

Table 1. Denavit–Hartenberg (D-H) parameter table for the ABB IRB 120 industrial robot. The table provides the standardized kinematic parameters for each joint i, including the joint angle Θ_i [°], the link offset d_i [mm], the link length a_i [mm], and the twist angle α_i [°]. These parameters define the robot's geometry and are essential for kinematic modeling and motion analysis.

Joint i	Θ_i [°]	d_i [mm]	a_i [mm]	α_i [°]
1	0	290	0	−90
2	−90	0	270	0
3	0	0	70	−90
4	0	302	0	90
5	0	0	0	−90
6	0	72	0	0

3.4. Measuring the Power Consumption of an Industrial Robot

Many recent studies have been published on the analysis and modeling of energy consumption of industrial robots [80,81]. Commonly, various components in robotic systems consume energy, such as controllers, cooling fans, motors, and friction joints of the robot [18]. In [82], the motor characteristics were analyzed by modeling the energy consumption of an ABB IRB 1200 robot based on the method of identifying parameters from power data rather than joint moments.

Our study adopts a qualitative research approach based on theoretical frameworks and previous research results, combining experimental and simulation methods to measure the energy consumption of robots. The main method is simulation in ABB RobotStudio 2024.1.1, which allows not only flexible modeling and analysis but also efficient testing of different robot configurations and trajectories.

Several approaches have been evaluated for the accurate measurement of energy consumption. The first approach involves measurement at the power supply of the robot

control unit, where measurement devices, such as wattmeters, are placed on the input power cable leading to the control unit. This configuration records the total power consumption of the robotic system before it is distributed to individual components such as motors and actuators, providing a comprehensive view of the overall energy efficiency of the system. In this method, the energy consumption is calculated as the product of power (in watts) and time (in hours), allowing for a simple calculation of the total system consumption over a period of time.

The second measurement approach involves the installation of metering devices at the input of each inverter, allowing the power consumption of the individual motors of the robot to be monitored. Each inverter powers a specific motor, allowing detailed power consumption analysis for individual robot components. This approach provides a detailed view of the operation of individual motors and is useful for diagnostics and energy optimization but requires a more complex installation that can affect the operation of the robot.

Due to the limitations of physical measurements, the authors decided to use simulations in ABB RobotStudio 2024.1.1, which allowed them to model and analyze the energy consumption without physical intervention in the system. The ABB RobotStudio 2024.1.1 software is based on accurate models of the robot, its components, and electrical characteristics, allowing highly accurate predictions of energy consumption. Simulation provides flexibility to test different types of robots that are not directly available, while reducing the cost and time required to install metering equipment. This simulation approach also allows testing of different operating conditions and robot trajectories, ensuring a high level of accuracy and reproducibility of results, which are key to optimizing energy efficiency.

As part of our research, a simulation model was created for several 6-axis robots in ABB RobotStudio 2024.1.1. The robots were simulated with no tools attached to obtain baseline energy consumption data under ideal, controlled conditions. Setting up the simulations in this way allowed us to accurately analyze the impact of different robot base positions on the robot's energy consumption and time efficiency, providing relevant data for further applications.

A module was used in the ABB RobotStudio 2024.1.1 software to measure the energy consumption of the robot joints and the entire manipulator [83]. The measurement of the electrical energy of these joints and the overall system can be implemented using energy sensors that estimate the consumption for a specific task. However, due to the limited availability and number of energy sensors, the strategy presented in this paper focused on an approach based on modeling the energy consumption of robotic manipulators.

The proposed framework provides an alternative framework for energy efficiency analysis without the need for physical sensors. For this reason, the research was limited to ABB robots.

Equipped with a virtual robot controller (VC) that is an exact copy of the software used on the physical robot controller, the ABB RobotStudio 2024.1.1 (RS) system enables realistic simulations and accurate estimates of the robot's energy consumption. This system provides the advantage of creating a digital twin, where the real robot controller in the production cell can connect to the virtual controller for offline programming and virtual commissioning. This integration allows flexible and efficient operations planning, testing, and optimization without the need to physically handle the robot.

The use of offline programming (OLP) allows accurate estimation of cycle time, peak power, and energy consumption, as well as verification of the feasibility of motion trajectories, which are key simulation outputs for a multi-objective optimization study. This approach is applicable in multiple phases—from planning to virtual commissioning to optimization of an existing robotic cell.

Technical specifications such as motor torque constants, torque characteristics, gear ratios, and efficiency are required to accurately calculate the resulting joint torques from measured current data. However, these technical specifications are restricted by ABB's policy of not allowing public dissemination. These limitations complicate direct analysis

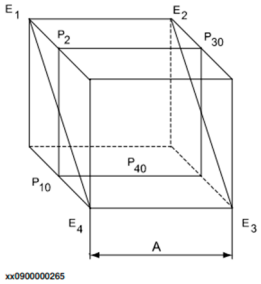
and calculation at the level of detailed mechanical parameters, but the RobotStudio 2024.1.1 ABB RobotStudio 2024.1.1 2024.1.1 system still provides sufficiently accurate results for practical optimization purposes.

Unfortunately, the electrical and mechanical parameters of the robot motors are not listed in their datasheets, but the datasheet for the ABB IRB 120 industrial robot defines a power consumption of 0.24 kW at its maximum load and maximum speed according to the ISO Cube trajectory [84] (Figure 3).

Path E1-E2-E3-E4 in the ISO Cube, maximum load.

Type of movement	Power consumption (kW)
ISO Cube Max. velocity	0.24 kW

Robot in 0 degree position	IRB 120
Brakes engaged	0.095 kW
Brakes disengaged	0.173 kW



Position	Description
A	250 mm

Figure 3. Power consumption analysis of the ABB IRB 120 industrial robot based on the ISO Cube standard under maximum load conditions. The table summarizes power consumption during different operating states: maximum velocity, stationary with brakes engaged, and stationary with brakes disengaged. The ISO Cube trajectory, illustrated in the diagram, defines the path *E1-E2-E3-E4* with a segment length A = 250 mm. These measurements provide insights into the robot's energy efficiency during dynamic and static operations [84].

Industrial robots usually consume alternating currents. The power flow of the trans-industrial robot is shown in Figure 4. The total energy is consumed by the control system and auxiliary components. The auxiliary components include a teaching trailer eye, an air compressor for the pneumatic gripper, fans to cool the control unit, and a computer to interact with the control system.

The performance of these auxiliary components is generally constant. The power delivered to the control system includes power dissipation (power consumption of AC/DC converters, DC bus, inverters, etc.), and the power delivered to the robot motors can be divided into two parts: one part is used to drive the motor, and the other part is the power dissipation of the motors [18].

The motors currently used in industrial robots are generally permanent magnet synchronous motors with high efficiency (83% to 92%). As can be seen from Equation (4), the power losses include copper, iron, mechanical, and stray losses and can be expressed as follows:

$$PP_{m,loss} = P_{Fe} + P_{copper} + P_F + P_{stray} \quad (4)$$

where $P_{m,loss}$ is the motor power loss, P_{Fe} is the iron loss, P_{copper} is the copper loss, P_F is the mechanical loss, and P_{stray} is the stray loss.

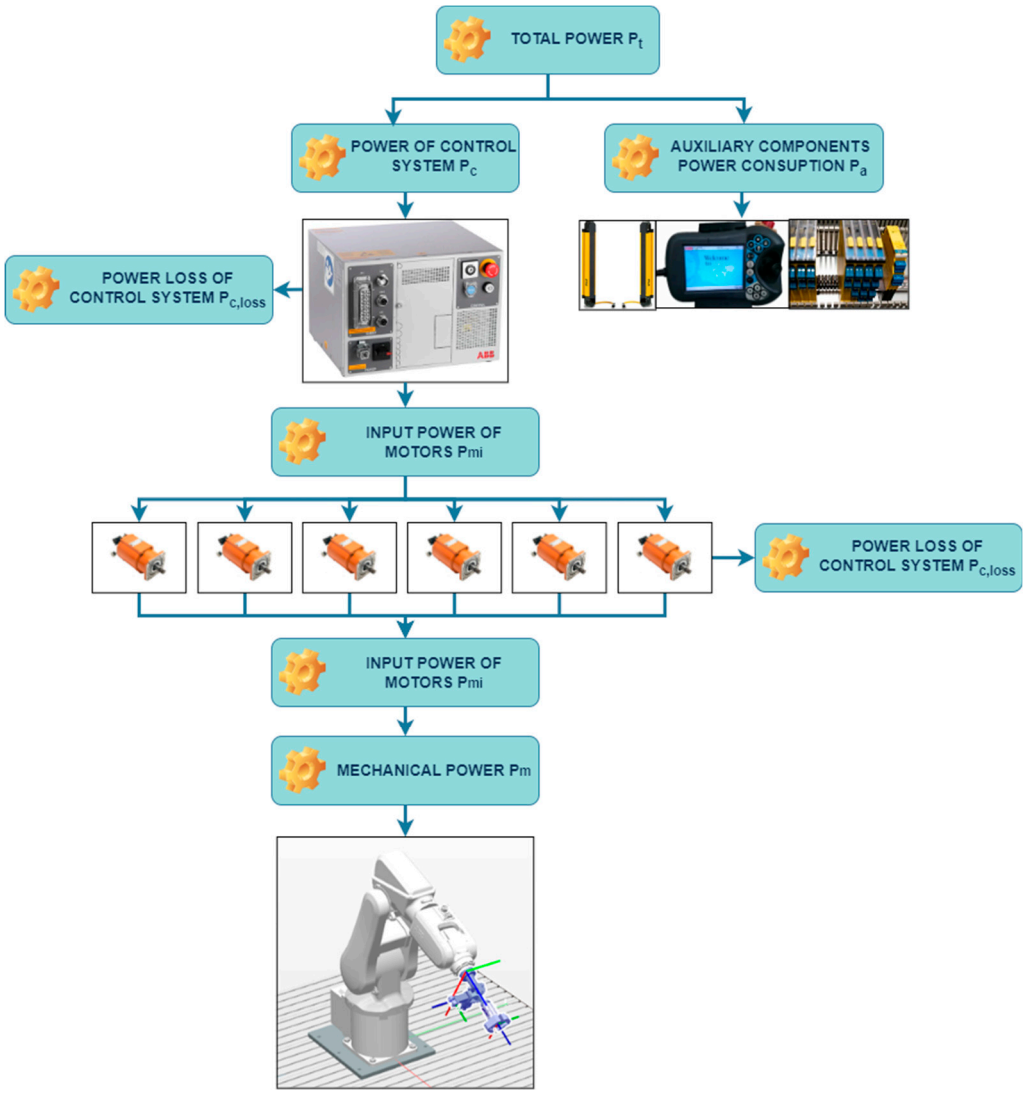


Figure 4. Energy flow of the ABB IRB 120 industrial robot, detailing the allocation and transformation of energy within the system. The total power input (P_t) is divided between the control system (P_c) and auxiliary components (P_a). Energy flows through the motors (P_{mi}), ultimately converting into mechanical power (P_m) at the end-effector. The diagram also highlights energy losses ($_{Pc,loss}$), providing a clear visualization of the system's energy efficiency.

3.5. Energy Consumption Model in ABB RobotStudio 2024.1.1

In order to build an energy consumption model for robotic operations, it is necessary to track the basic electrical quantities that determine the amount of energy required for the robot's motion tasks. For simulations in the ABB RobotStudio 2024.1.1 environment, authors focus on variables such as voltage (V) and current (I), which form the basis for calculating the robot's eye power (P). This power represents the instantaneous energy consumption of the robot, which can be monitored in real time. The total energy consumption (E) is then determined by integrating the power over time, thus obtaining the cumulative value of the energy required for task execution.

When monitoring the energy requirements, the energy consumption at the level of individual joints is also analyzed. Each joint can exhibit positive power when accelerating or negative power when decelerating, thus creating the possibility of energy recuperation between joints. If one joint decelerates and another accelerates, energy from the decelerating joint can be efficiently transferred to the accelerating joint, reducing the overall energy

demand. This mechanism helps to identify energy-intensive sections of the trajectory, allowing for better optimization of energy consumption.

In addition to the instantaneous power, authors also track variables such as Total Motor Power, which represents the instantaneous power of the individual robot motors, and Total Motor Energy, which provides the cumulative power consumption of the motors. These variables are influenced by the dynamic simulation parameters such as robot speed, acceleration, and load. The energy consumption calculation also includes the dynamic aspects of the robot motion, where the velocity and acceleration of the joints affect the dynamic load on the motors and directly affect the power input of the robot.

Setting up the simulation involves defining basic robot motion parameters such as speed and load, while the simulation tracks instantaneous and cumulative energy consumption throughout the operation. The resulting data allow the analysis of energy peaks and the identification of sections where operating parameters can be optimized without the need for physical intervention in the robot.

It should be noted that the exact internal calculation data used by ABB RobotStudio 2024.1.1 to simulate energy consumption is not publicly available due to proprietary rights. For example, the exact algorithms that the software uses to simulate dynamic factors such as torque, power factor, and motor efficiency are not available for detailed analysis. For this reason, the authors conceived the model based on basic physics and electrical engineering principles, using common formulas to calculate power, torque, and power consumption.

Despite these limitations, the simulations provide a sufficiently accurate picture of energy requirements for workplace planning and optimization. This energy consumption model is therefore applicable to identify optimal conditions for energy-efficient robot operations and is designed to be reproducible and applicable to different types of robotic systems.

For each power consumption measurement, the authors followed the following procedure in ABB RobotStudio 2024.1.1. First, a simulation for the robot model was set up, defining the tasks the robot was to perform and specifying simulation parameters such as speed, acceleration, and load. The authors then activated the energy measurement tool, which monitors the instantaneous power output of each motor and records the cumulative energy consumption throughout the simulation.

The simulation was run, and data on instantaneous power (in units of watts) and cumulative energy (in units of watt-hours) were collected in real time. These notes allowed the authors to track the dependencies between the robot's motion parameters and its energy consumption, and the results were analyzed using graphs and tabs generated in ABB RobotStudio 2024.1.1. This procedure gave the authors an overview of the energy requirements of the different trajectories and tasks of the robot.

However, when measuring the power consumption of the ABB IRB 120 robot, the authors extended the procedure to include the connection of the physical robot. The robot was connected to a system that online monitored its actual power consumption during the execution of specified tasks. Then, they compared the results from the real measurements with the simulation data to verify the accuracy of the simulated values and to confirm that the simulation accurately reflected the energy requirements of the real robot system. This approach allowed the authors not only to verify the accuracy of the simulations but also to optimize the energy consumption based on real data.

This procedure is generally valid according to the manufacturer's instructions for ABB RobotStudio 2024.1.1. The manufacturer recommends this procedure for monitoring and analyzing the energy consumption of robotic systems through simulation. The authors followed these recommendations for each measurement, which allowed us to accurately model the energy requirements of the robotic operations and monitor both instantaneous and cumulative power.

In the case of the ABB IRB 120 robot, they complemented this procedure with a direct connection to the physical robot in order to compare the real measured values with the simulation and to verify the accuracy of the data obtained from the simulation.

3.6. Consumption Comparison Between Real and Simulated Robots

The parametric trajectories were uploaded to the robot controller at a 50% range in the X-axis and a height range from 10% to 70% of the highest reachable point from the robot base. To increase the difficulty of trajectory execution, the following parameters were selected: maximum velocity (v_{max}), maximum acceleration (a_{max}), zone 0 (z0), and payload 0.

The last parameter indicates that the robot had no tool clamping; thus, the load of additional mass on the last axis of the robot was zero.

Due to the space constraints of the site, the robot could not perform trajectories at 20% even if it was structurally capable of doing so. The experimental measurement model is shown in Figure 5.

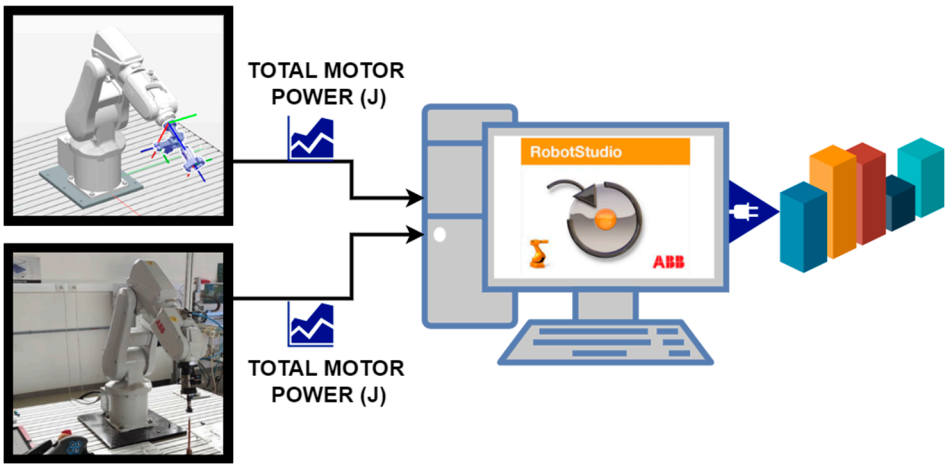


Figure 5. Energy consumption measurement framework for the ABB IRB 120 industrial robot. The model integrates both physical measurements from the real robotic workstation and simulated data from ABB RobotStudio 2024.1.1. The total motor power consumption (*J*) is calculated and analyzed in both environments, enabling a comprehensive evaluation of energy usage. This approach facilitates validation and comparison of real-world and simulated energy consumption, supporting the optimization of robotic processes.

The figure shows the process of measuring and analyzing the robot's energy consumption using simulation in ABB RobotStudio 2024.1.1. The upper part shows the robot model in the simulation and the lower part shows the actual physical robot ABB IRB 120. Both systems are connected to a computer running ABB RobotStudio 2024.1.1 software, where the Total Motor Power variable (in joules) is monitored to represent the instantaneous power consumed by the robot's motors.

The authors proceeded by running a simulation in ABB RobotStudio 2024.1.1 and measuring the energy consumption for different robot tasks. At the same time, they connected a real ABB IRB 120 robot and monitored its energy consumption online during the same tasks. In this way, they compared the simulated values with the actual measured data, which allowed them to verify the accuracy of the simulation and obtain detailed data for further analysis and optimization of the robot workstation. Figure 6 plots the measured power consumption values of the real robot (blue) versus the measured values from the simulation (orange). In the graph, a marked difference between the measured values can be observed, but the shape of the graph remains unchanged.

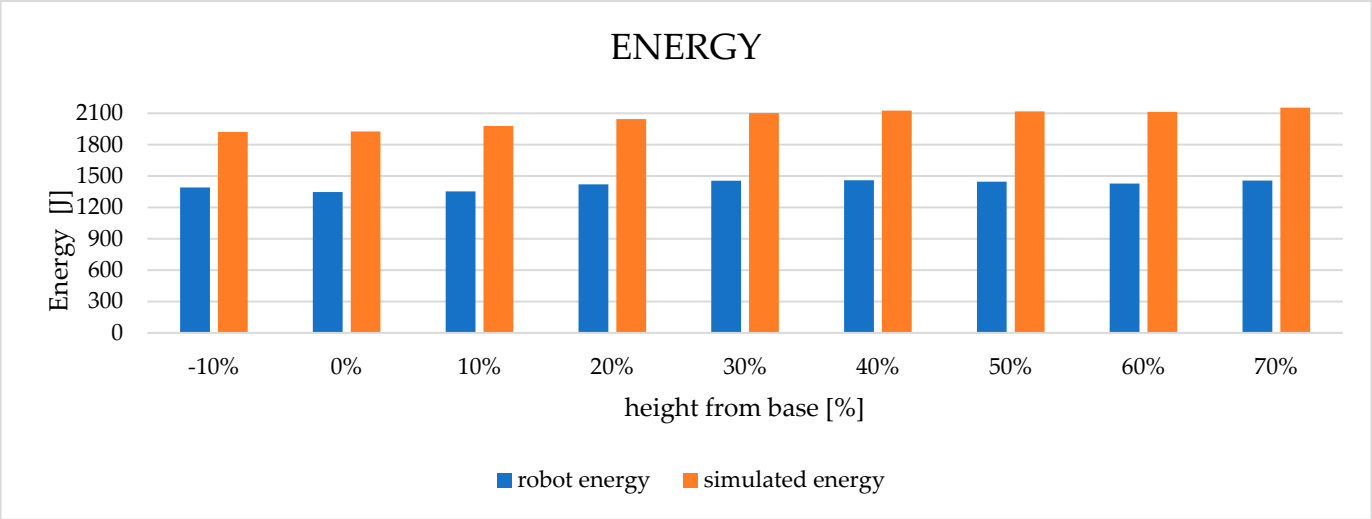


Figure 6. Measured energy consumption of the ABB IRB 120 robot at varying heights from the base, expressed as a percentage. The blue bars represent the real energy consumption measured during physical experiments, while the orange bars represent the simulated energy consumption.

It is also possible to see different power consumptions depending on the altitude at which the trajectory was performed.

Figure 7 shows a graph that shows the time taken by the real robot (blue) and the robot in the simulation (orange) to perform the trajectory.

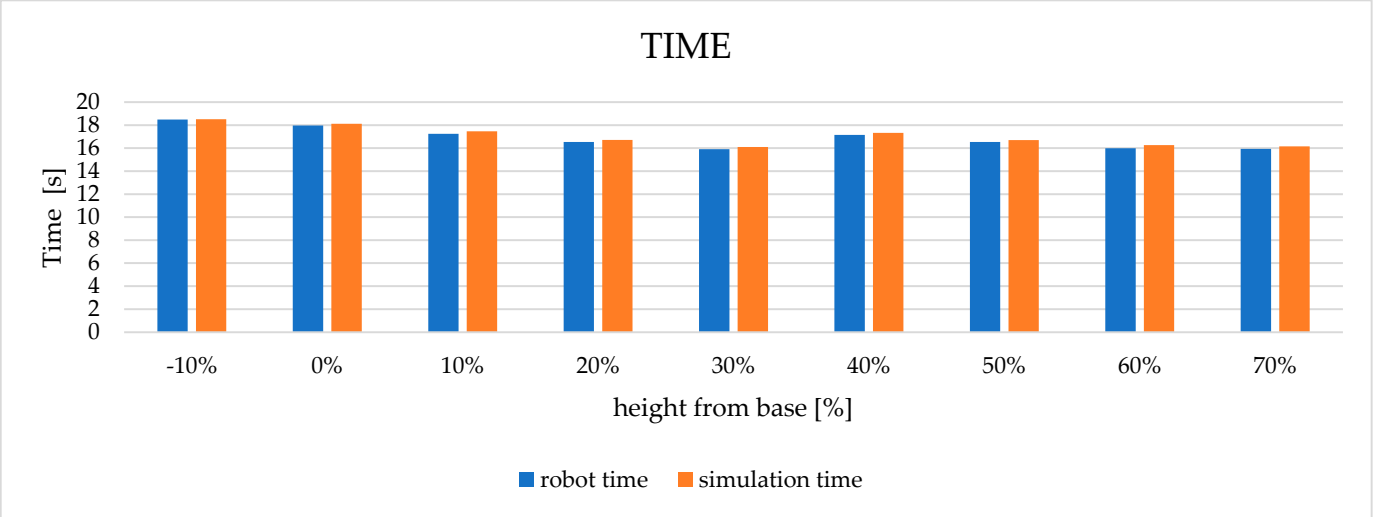


Figure 7. Comparison of measured execution times for the ABB IRB 120 robot across different heights relative to the base, expressed as a percentage. The chart illustrates the time required for task execution in both the physical robot (blue) and the simulated environment (orange). The results highlight the consistency between real-world and simulated performance, with minor variations observed at different height levels, offering insights into the accuracy of simulation models for time estimation.

From the graph, it can be seen that the measured values are almost identical, and the differences are only minimal. As in the previous case, it is possible to note the different durations of the trajectory depending on the height at which the trajectory was performed.

An important factor in monitoring the actual power consumption of a robot is that the total motor power represents the mechanical consumed power, not the power that is fed into the control cabinet from the power grid. Thus, power consumed by the control

cabinet is excluded. In the case of a virtual robot, the measured throw notes are based on the nominal robot under typical conditions.

For the virtual robot, the measured torque was based on the torque under real conditions. For a real robot, the measured motor power value depends on various factors, such as the robot temperature and the length of the cables. The manufacturer reports a 30% increase in the power consumption of the virtual robot compared to reality due to the inclusion of the control cabinet power consumption in the simulation.

The measured values from the real and simulated robots are listed in Tables 2 and 3, where the differences between the measured values are compared and their differences are calculated. For the measured time values, the difference between the simulation and the real robot was at the decimal place level, which was approximately a 1% difference.

Table 2. Comparison of execution times for the ABB IRB 120 robot between real-world measurements and simulation at various heights. The table shows measured robot time, simulation time, and their differences, with deviations mostly within 1–2%, confirming the simulation's accuracy for predicting real-world performance.

Height	Robot Time [s]	Simulation Time [s]	Difference	
−10%	18,483	18,516	0.033	0%
0%	1797	18,113	0.143	1%
10%	17,244	17,458	0.214	1%
20%	16,533	16,716	0.183	1%
30%	15,911	16,094	0.183	1%
40%	17,155	17,328	0.173	1%
50%	16,533	16,699	0.166	1%
60%	15,991	16,259	0.268	2%
70%	1594	16,152	0.212	1%

Table 3. Energy consumption comparison for the ABB IRB 120 robot in real and simulated environments across various heights. The table shows measured robot energy, simulated energy, and the differences, highlighting a consistent offset of 28–32% between the two environments.

Height	Robot Energy [J]	Simulated Energy [J]	Difference	
−10%	139,002	1,921,278	531,258	28%
0%	1,346,399	1,926,106	579,707	30%
10%	1,352,988	1,978,026	625,038	32%
20%	1,419,459	2,044,792	625,333	31%
30%	1,454,371	2,101,298	646,927	31%
40%	1,459,061	2,125,478	666,417	31%
50%	1,445,096	2,118,259	673,163	32%
60%	1,427,914	2,112,532	684,618	32%
70%	1,456,751	2,153,973	697,222	32%

When measuring electricity consumption, it is already possible to notice a difference in almost one-third between the measured values. After recalculating the difference in the measured values, it can be concluded that the values measured purely in the simulation environment are inflated by about 31%, as stated by the manufacturer; thus, the measured electricity consumption values can be considered relevant.

3.7. Selection of the Sample of Industrial Robots to Be Studied

To develop a methodology suitable for all angular industrial robots, a group of seven ABB robots with different maximum reaches (0.5–4.2 m) were selected.

In addition, some selected industrial robots had different design features, such as parallelograms. These industrial robots were ABB IRB 120, ABB IRB 1200, ABB IRB 2600, ABB IRB 4400, ABB IRB 5720, ABB IRB 6700, and ABB IRB 8700.

Preliminary time measurements were made using a constructed parametric trajectory, which is described later. The points of the trajectory had a constant distance in the X-axis (50% of the maximum range), and the positions of the points in the Z-axis were gradually changed to determine the positions of the minimum and maximum time.

From the plot of the preliminary time measurements (Figure 8), the greater the reach of the selected industrial robot, the lower the least time-consuming position. In most cases, the most time-consuming positions of the selected industrial robots were at or below the level of the robot base.

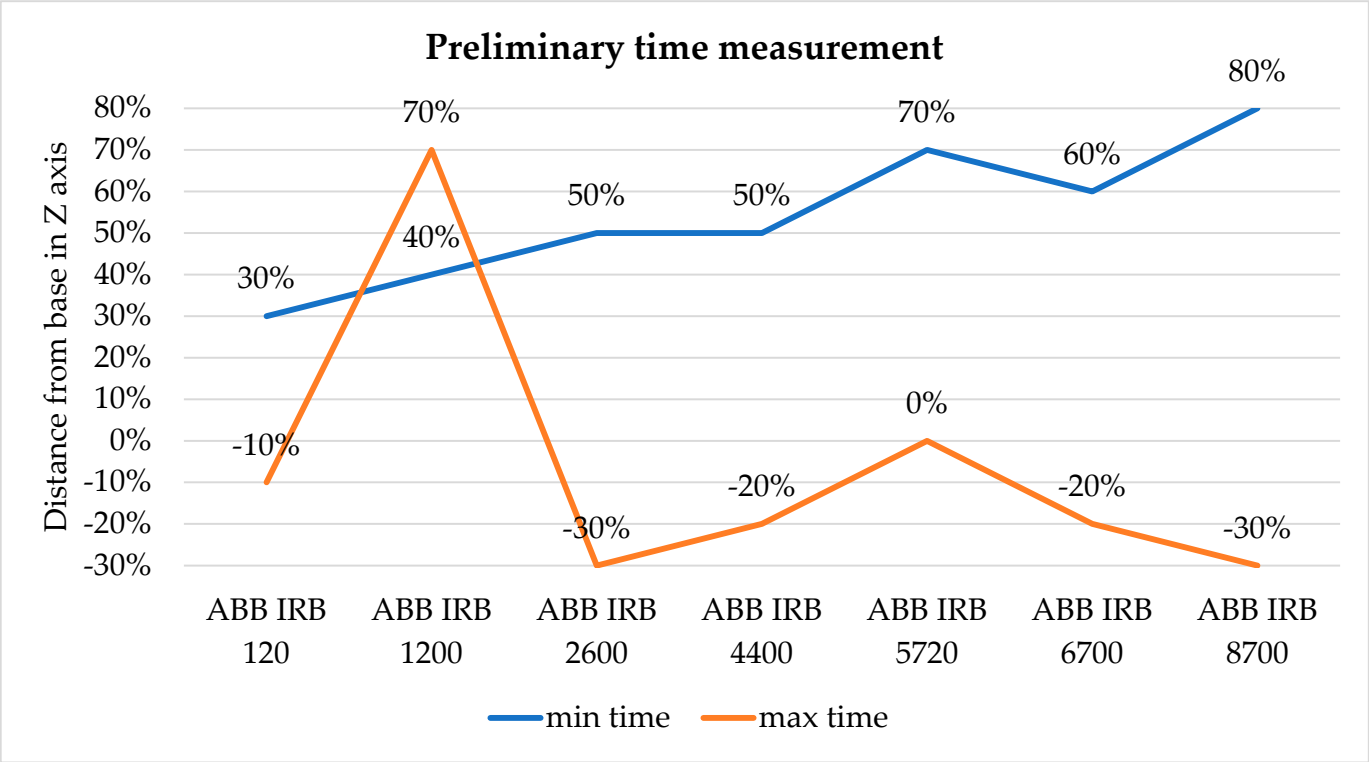


Figure 8. Preliminary time measurements for various ABB robot models. The graph compares the minimum and maximum task execution times as a function of the distance from the base in the Z-axis, expressed as percentages. The results illustrate variations in performance across different robot models, highlighting the impact of distance on execution time.

The only exception was the ABB IRB 1200 industrial robot, which had the highest time demand of 70% of its maximum range from the robot base.

However, preliminary measurements of power consumption (Figure 9) revealed an uneven distribution of the working height with respect to the minimum and maximum power consumption. Most of the selected industrial robots have a minimum power consumption position at or below the base level.

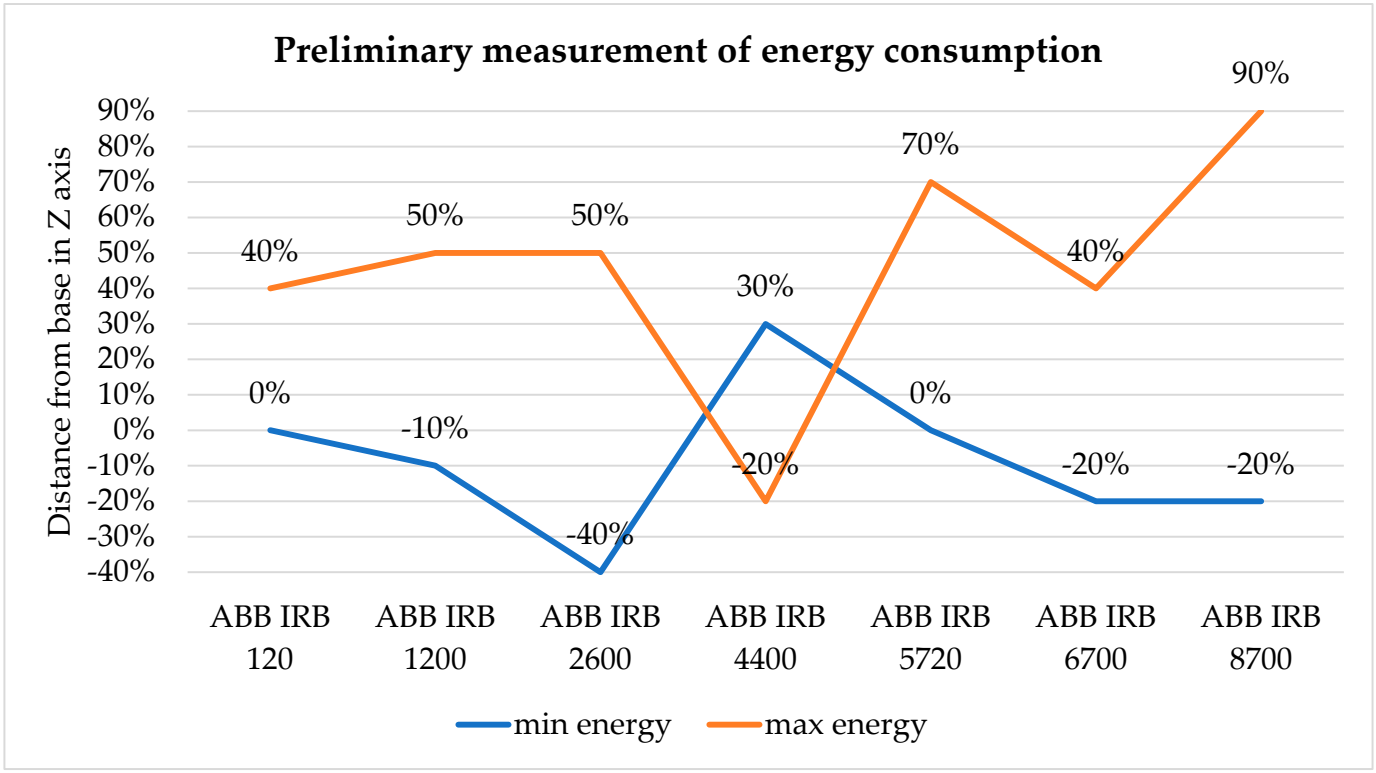


Figure 9. Preliminary energy consumption measurements for various ABB robot models. The graph compares the minimum and maximum energy consumption as a function of the distance from the base in the Z-axis, expressed as percentages. The results demonstrate significant variations in energy usage across robot models, highlighting the influence of operational height and model-specific factors on energy efficiency.

In most cases, the maximum energy demand of the selected robots exceeds 40% of its maximum range at the distance from the base. However, the ABB IRB 4400 industrial robot had the minimum and maximum energy intensity positions in reverse order. The maximum energy intensity position is below the robot base and the minimum energy intensity position is above the robot base.

The different positions may be due to the different designs of the industrial robot, as this robot has a parallelogram.

Considering the previously measured values of the minimum and maximum energy and time consumption positions, it was necessary to reconsider the selection of industrial robots suitable for the developed methodology. In order to determine in which position the selected industrial robots should work, the measured values of both minimum and maximum time–energy intensity must be approximately at the same level.

Therefore, industrial robots that are structurally similar, and have a range of 2.3–3.1 m and a payload capacity of 90–270 kg, were subsequently selected. None of the selected industrial robots had a parallelogram. The selected industrial robot models were ABB IRB 5710, ABB IRB 5720, ABB IRB 6710, ABB IRB 6720, ABB IRB 6730, and ABB IRB 6740.

Another reason for choosing these industrial robots is their frequent use in the automotive industry and their numerous deployments in body shops and assembly.

Based on the measured values, it is possible to determine which distance and height from the base are the least and most energy- and time-consuming for the selected industrial robots.

3.8. Parametric Trajectory Creation

The trajectory consists of 19 points arranged in a semicircle, with a 10° rotation between each point. This approach was chosen because the semicircular motion over a range of

180 degrees with a uniform 10-degree spacing allows an accurate analysis of the robot's motion and its impact on energy consumption. The points are spaced such that each point of the trajectory is equidistant from the robot base in the X and Z axes, ensuring consistent spacing and eliminating the impact of position change on energy consumption. The points in the left half of the semicircle are mirrored from the points in the right half, ensuring trajectory symmetry.

This trajectory simulates a pick-and-place operation. The first point is directly in front of the robot with a 0° offset, which represents the starting position. The second point is offset by +90°, where the robot performs the "Pick" operation. The robot then moves back to the 0° position where it places the object ("Place"). This pattern is repeated with a gradual decrease in rotation—for example, to +80° and back to 0°, until the last combination of −90° and 0°. In this way, the energy and time requirements for each trajectory step can be thoroughly verified.

The parametric trajectory in Figure 10 was created in Process Simulate and is applicable to all selected robots, thereby allowing consistent testing and comparison of the energy efficiency of different models.

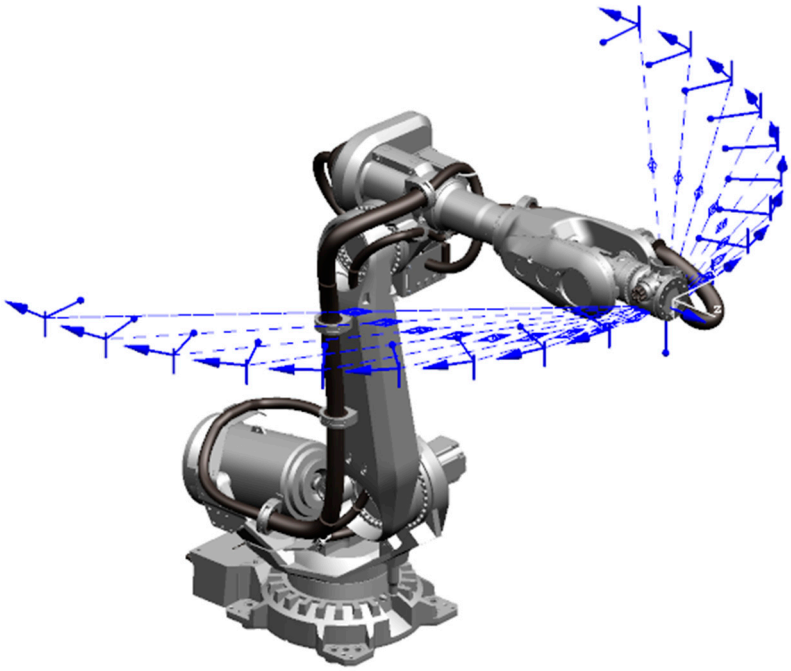


Figure 10. Visualization of a parametric trajectory for a robotic manipulator. The figure illustrates the predefined path in 3D space, represented by a sequence of parametric points and directional vectors, demonstrating the robot's capability to follow complex trajectories with precision and flexibility. The blue arrows represent the working point in space and its position in the right-hand coordinate system.

To verify the electrical consumption and trajectory execution time with respect to the position of the working object on the X and Z axes in percentages, trajectories were created for the robots in increments of 10% of the maximum achievable height and distance from the robot's base. As an example, the ABB IRB 6740 robot has a maximum Z-axis reach of 3597.5 mm from the base.

From this value, it can be determined that 10% of the maximum achievable height is 359.75 mm. This value determines the spacing between the trajectories along the Z-axis. The same procedure is applied to the X-axis spacing. The maximum achievable distance is 3162.84 mm; thus, the trajectory radius changes by 316.284 mm.

Due to the different limits of the designs and the lengths of the individual arm parts, the number of trajectories was different for each robot. The number of trajectories for a

single robot ranged from 19 (ABB IRB 6740) to 42 (ABB IRB 5720). The experimental model for the measurement is shown in Figure 11.

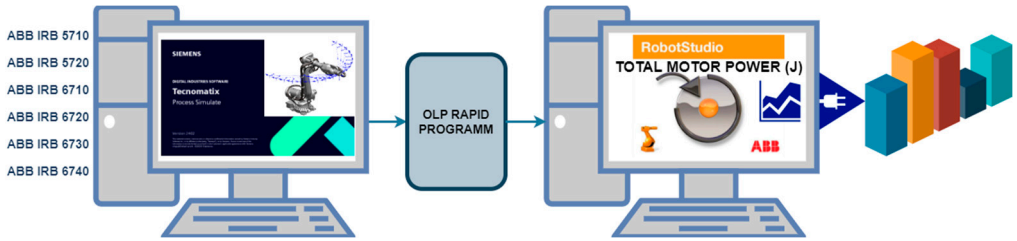


Figure 11. Experimental model for energy and time measurement using co-simulation. The model connects Siemens Tecnomatix Process Simulate 2402 and ABB RobotStudio 2024.1.1 to analyze motor power consumption and optimize robotic performance.

The figure shows the process of simulation and analysis of energy consumption for different ABB robot models (IRB 5710, IRB 5720, IRB 6710, IRB 6720, IRB 6730, and IRB 6740) using the Siemens Tecnomatix Process Simulate 2402 and ABB RobotStudio 2024.1.1 software tools.

On the left side is a computer with Siemens Tecnomatix Process Simulate 2402 software, where motions for different robot models are created and simulated. An OLP (Offline Programming) program in RAPID format is generated from Tecnomatix, and it defines the movements and tasks of the robot.

In the next step, the OLP RAPID program is loaded into ABB RobotStudio 2024.1.1 (shown on the right), where the Total Motor Power variable (in joules) is monitored. The proposed tool allows real-time measurement of the robots' power consumption during the simulation. The resulting power consumption data are then analyzed, providing insights into the energy consumption of individual movements and optimizing the power consumption for different robot models.

In this way, the authors simulated and evaluated the energy requirements of the robots based on the generated motion trajectories and analyzed the possibility of optimizing the energy consumption for each robot model.

Another criterion for the generated trajectory is for the robot to maximize the utilization of its axes when reaching points, thus increasing the energy and time requirements of the trajectory due to more accurate measurements. Since the trajectory points have a constant distance value along the X and Z axes from the robot base, axes 2 and 3 are less utilized in this type of trajectory. These axes primarily define the position of the point in space (the distance of the working point in the Z-axis from the base), which is invariant in this case.

Therefore, axes 4, 5, and 6, which define the orientation of the work object in space, must be more stressed. Since all points on the right or left sides had the same rotation in space, it was necessary to adjust the robot configuration at every other point. This was performed to maximize the use of axes 4, 5, and 6.

Because the thesis was designed in such a way that the results could be used in practice, the number of trajectories was reduced to 173 from the original 274 trajectories created for the research. This is because in practice, when creating trajectories, it is necessary to limit the range of individual robot joints by 5% to 10% (depending on the requirements of the specific company).

This limitation was attributed to the inaccurate placement of objects, as it was not possible to build a 1:1 robotic cell for the simulation. Therefore, any trajectory in which the robot had to exceed the given constraint to reach the desired point cannot be used. In this study, a constraint of 5% was chosen. The generated trajectories were exported and modified to match the appropriate formatting of ABB's programming language for RAPID, an industrial robot.

The graph in Figure 12 shows the utilization of axes (their rotation angle) 1, 4, 5, and 6 during trajectory execution for the ABB IRB 6740 robot at 70% height and 50% distance from the robot base. At the bottom of Figure 12, the total utilization of all axes (hatched) and the minimum and maximum rotation values of a given axis are indicated. The pink region represents the 5% axis rotation limit, which cannot be exceeded.

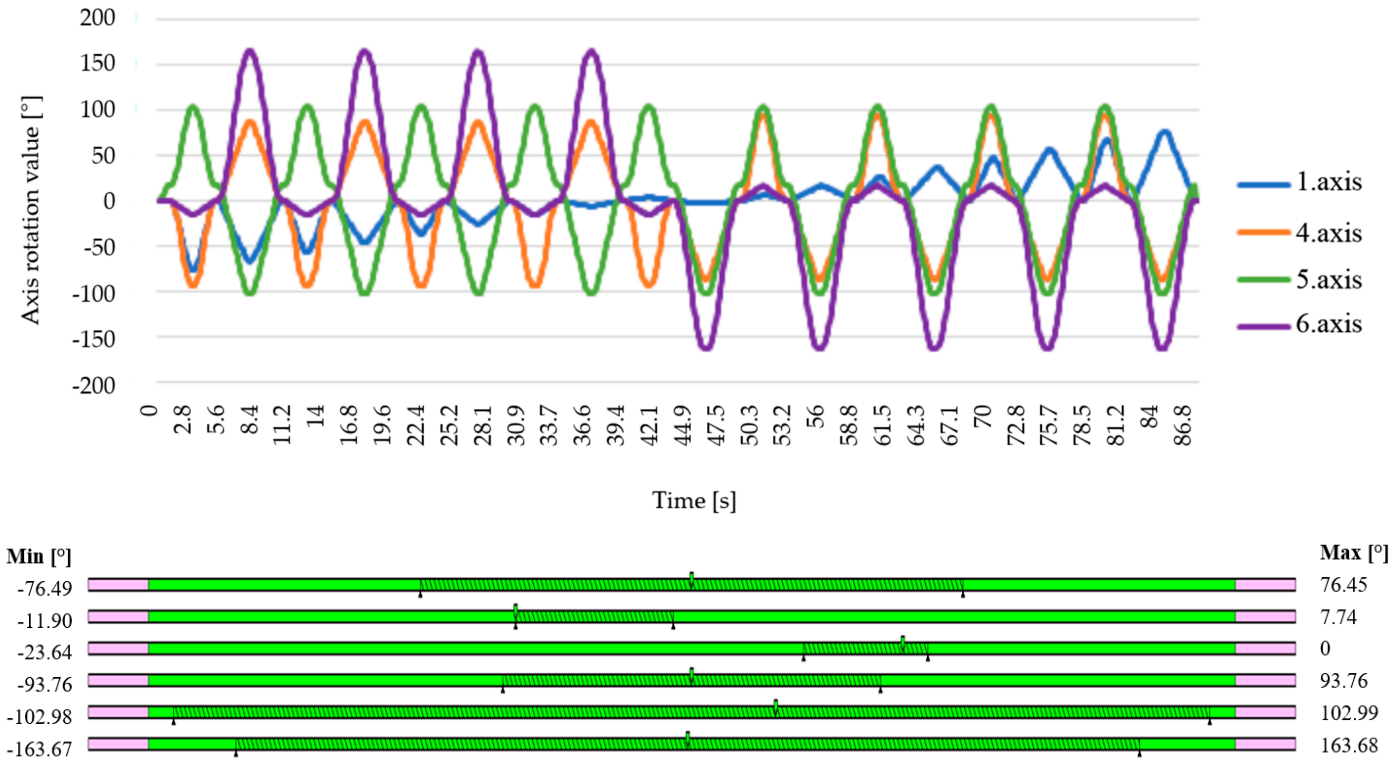


Figure 12. Robot arm axis utilization over time. The upper graph shows the dynamic rotation values of the robot's 1st, 4th, 5th, and 6th axes, with each axis represented by a specific color: blue for the 1st axis, orange for the 4th axis, green for the 5th axis, and purple for the 6th axis. The lower chart illustrates the minimum and maximum rotation values of each axis. Green bars represent the utilized range of motion, while pink segments indicate restricted zones where the robot is not allowed to operate, as these areas are undesirable for the robot's motion. Black lines mark the recorded minimum and maximum rotation values, providing a clear visualization of axis utilization and limitations.

4. Results

Based on experimental measurements, simulations of the electric energy and time consumption of the ABB IRB 5710, ABB IRB 5720, ABB IRB 6710, ABB IRB 6720, ABB IRB 6730, and ABB IRB 6740 industrial robots were performed. A parametrically adjustable trajectory consists of 19 points, which are uniformly distributed around the industrial robot.

All trajectory points were separated by a constant distance along the X and Z axes from the base of the industrial robot. To illustrate the different energy and time requirements with respect to trajectory position, the trajectory was executed at different distances and heights from the base. Due to the different design parameters and limits of each type of industrial robot, the number of measurements for each robot is different. An additional constraint was applied to the measurements, which limited the motion of each joint of the industrial robot by 5° from its maximum use.

This additional constraint was used because of the practical application and eliminates the need to limit the range of individual workstation positions due to inaccurate placement of the components of the actual workstation relative to the position of the components in the simulation.

In total, 274 trajectories were evaluated. After applying the constraint, the number of suitable trajectories was reduced to 173. The ABB IRB 5710 industrial robot could perform 44 trajectories, of which 29 were suitable. The ABB IRB 5720 robot performed 59 trajectories, of which 40 were suitable. The ABB IRB 6710 robot executed 41 trajectories, 20 of which were in violation of the specified conditions. The ABB IRB 6720 robot executed 50 trajectories, of which 21 exceeded the constraint. The ABB IRB 6730 robot performed 44 trajectories, and 20 of them did not meet the constraints. The last cross-sensory robot, ABB IRB 6740, performed 36 trajectories.

There were 17 non-compliant trajectories. Each of these trajectories was measured 10 times at every possible height and distance from the base that the industrial robot could perform. The resulting values of the measured power and time consumption were then averaged and entered into tables, in which it was possible to determine the effect of power and time consumption on the working position of each industrial robot.

Another measurement was performed to verify the reliability of the measured values of the electrical consumption and trajectory time in the ABB RobotStudio 2024.1.1 simulation software. The measurements were performed by comparing the measured values of the power consumption and time of the parametric trajectory performed at different heights and distances in the simulation and real environments using the ABB IRB 120 industrial robot. The deviation of the measured values of the energy consumption of the executed trajectory in the simulation environment with respect to the real industrial robot shows an increase in the electrical consumption by 31% on average.

This difference in the measured values is due to different approaches to measuring the power consumption in the simulation and in the real device. While the measured value of the real industrial robot reads only the values from the actual industrial robot, the measured values in the simulation include the power consumption of the controller in the total power consumption. The manufacturer states that the difference between the measured values in the simulation and the real robot should be approximately 30% due to the reason mentioned above, which indicates that the measured values of power consumption are correct.

The measured time values were almost identical when comparing the simulation values and the actual measured time of the industrial robot. In this case, the deviation between the measured values was approximately 1%.

Figure 13 plots the positions of the minimum and maximum power consumptions of the selected industrial robots relative to the baseline.

The positions with the lowest energy intensity are indicated in green, and the positions with the highest energy intensity for each selected industrial robot are indicated in red.

The selected industrial robots were of similar design; thus, some marked positions were shared between the robots. From the graph, it can be seen that the positions with the lowest consumption are around 50% on the X-axis and range from +10% to 40% depending on the robot used.

At the highest energy intensity values, the position dispersion is larger. Nevertheless, some robots share a percentage position in space. As an example, industrial robots with ABB IRB 5720, 6710, and 6730 have the highest power consumption at X 50% and Z 80%, whereas robots 6720 and 6740 have the highest consumption at X 80% and Z 40%. The ABB IRB 6710 and 6740 industrial robots had the lowest consumption points, X 50% and Z 10%. The other positions were different for the selected robots.

The positions in the X and Z axes with the lowest and highest power consumption were then averaged, and based on these calculations, the optimal working position for the selected group of robots was determined.

From the graph in Figure 14, it can be concluded that the optimal distance for the lowest energy intensity on the X-axis lies at 48% from the base of its maximum range. For the Z-axis, the optimal working position was located at 10% of the maximum range below the robot's base. In contrast, the position at 63% of the maximum range on the X and Z axes is the most energy-intensive.

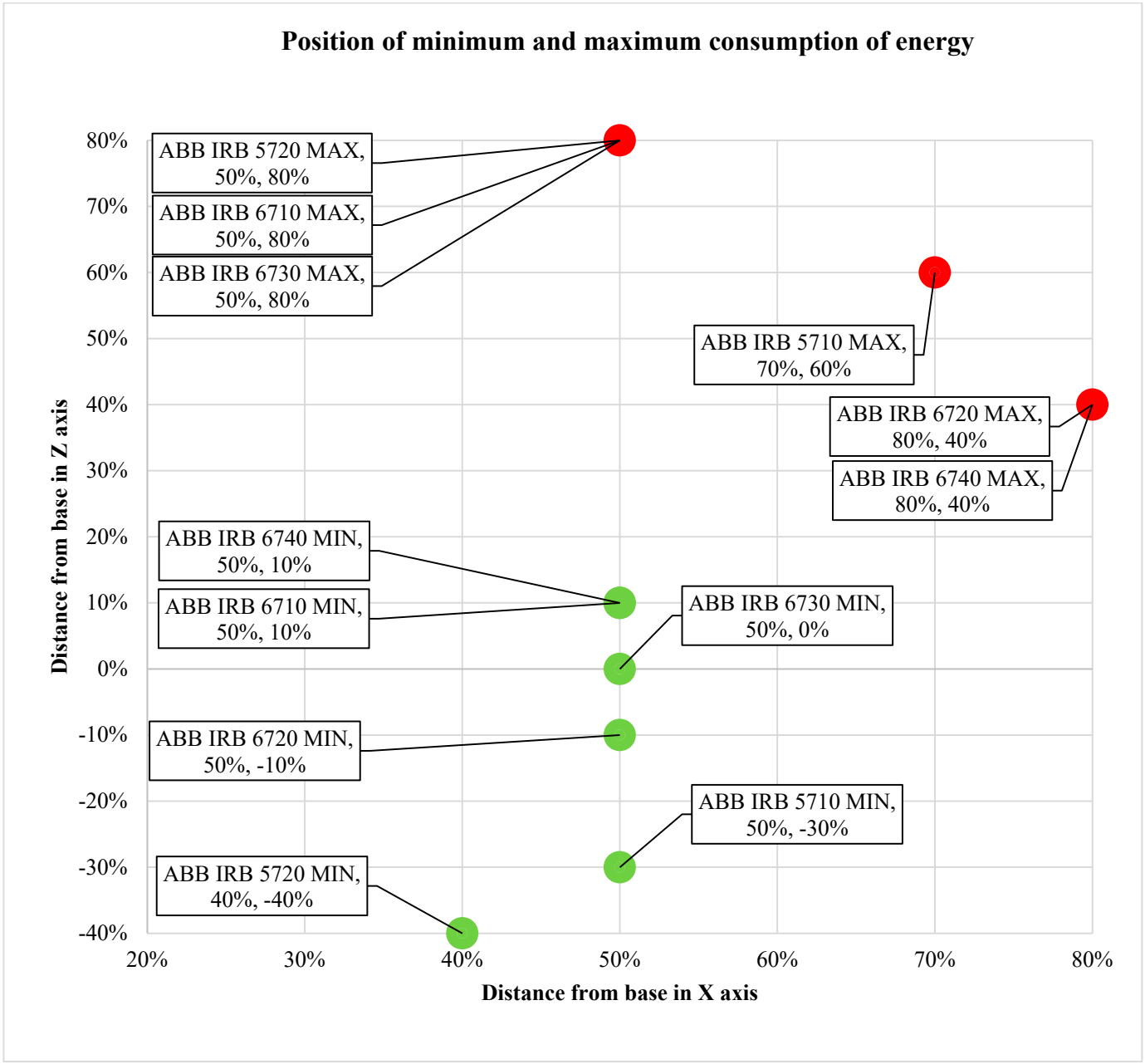


Figure 13. Positions of minimum and maximum electricity consumption for various ABB robot models. The chart plots the positions in the X and Z axes where energy consumption peaks (marked in red) or is minimized (marked in green). This visualization highlights the relationship between the robot's spatial position and its energy efficiency, providing insights into optimal positioning for energy-saving operations.

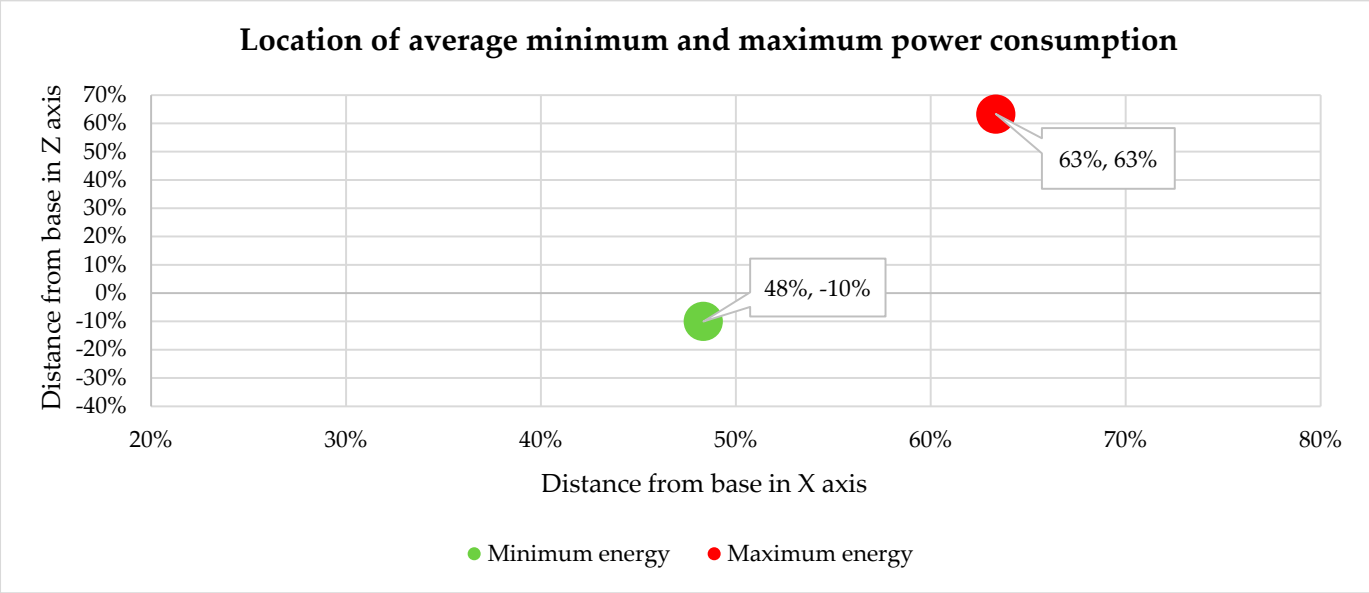


Figure 14. Locations of the average minimum and maximum electricity consumption. The chart identifies the spatial positions of average minimum (green) and maximum (red) energy consumption in the X and Z axes, illustrating the relationship between the robot's placement and its energy efficiency. These findings can guide optimal positioning for energy-intensive operations.

Figure 15 shows the least and most time-consuming positions of each selected robot. The green shows the least time-consuming positions, and the red shows the most time-consuming positions.

As in the previous case, the design similarity made some positions equally time-consuming for different robots. As an example, the common points with the lowest time complexity are the ABB IRB 6710, 6720, 6730, and 6740 robots. This position is at 50% in the X coordinates and 70% in the Z coordinates. Only the ABB IRB 6710 and 6740 robots had the common position with the highest time complexity. This position is in X = 60% and Z = 0% coordinates.

The coordinates with maximum and minimum energy intensities were averaged, and the distributions are shown in Figure 16. Here, it can be seen that, while for the energy intensity, the lowest average value was below the robot's base level, the location of the lowest time intensity was well above the base level. Specifically, it occurs at X 50% and Z 70%. The opposite is also true for cases with maximum time intensity. The position of maximum time difficulty was at coordinates X 63% and Z 3%.

In conclusion, it can be concluded that for the selected industrial robots, the least energy-consuming position was at X 48% and Z 10 coordinates. The position with the highest average power consumption was located at X 63% and Z 63% coordinates.

The time complexity is in the opposite position; thus, the lowest average value of the time required to perform the trajectory is found in coordinates X 50% and Z 70%, and the highest average value of the energy complexity is found in coordinates X 63% and Z 3%.

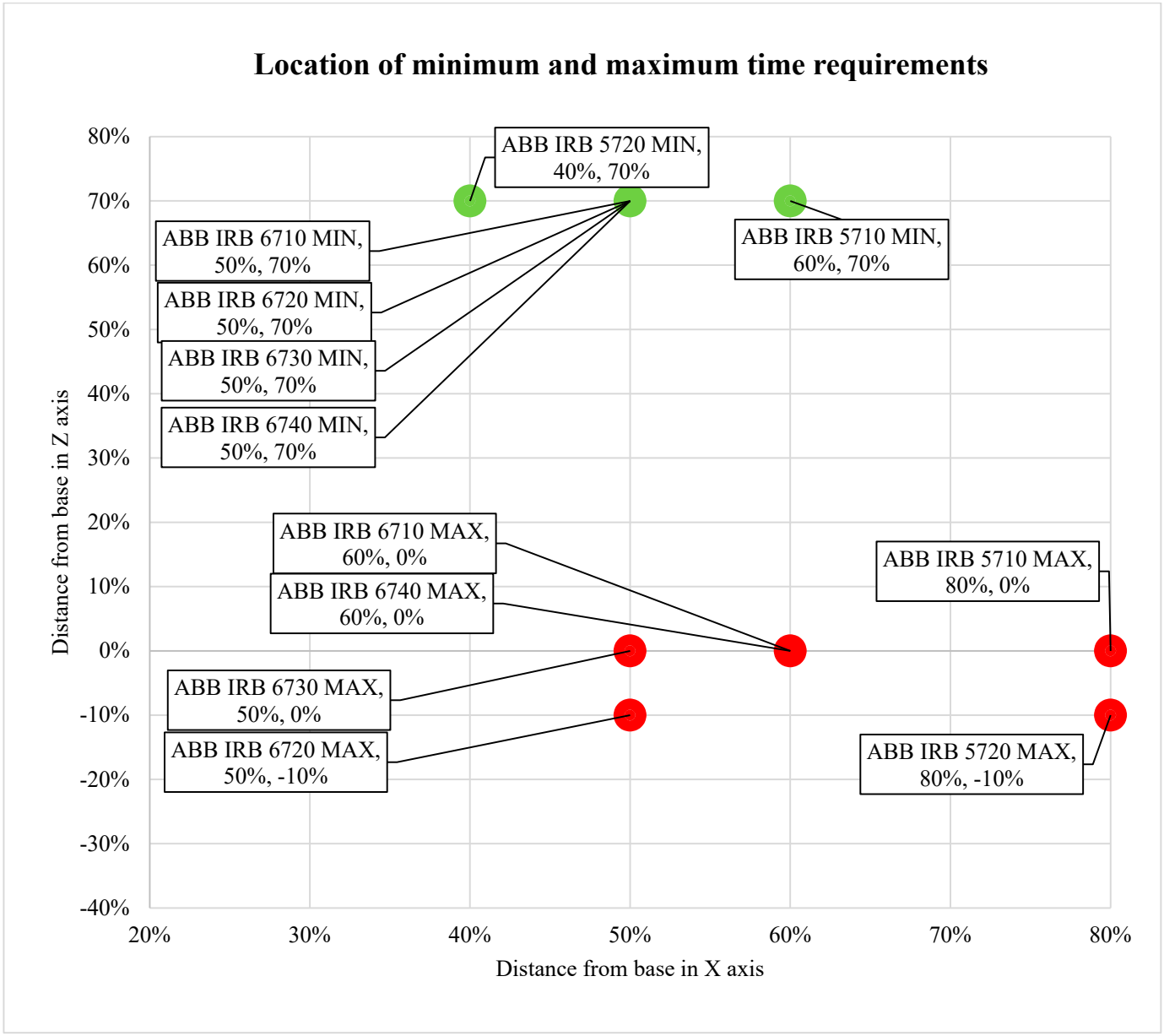


Figure 15. Positions of minimum and maximum time requirements for various ABB robot models. The chart depicts the locations in the X and Z axes where the time demand is minimized (green markers) and maximized (red markers). The analysis reveals spatial dependencies of operational time, offering insights for optimizing robot placement to reduce time requirements in specific tasks.

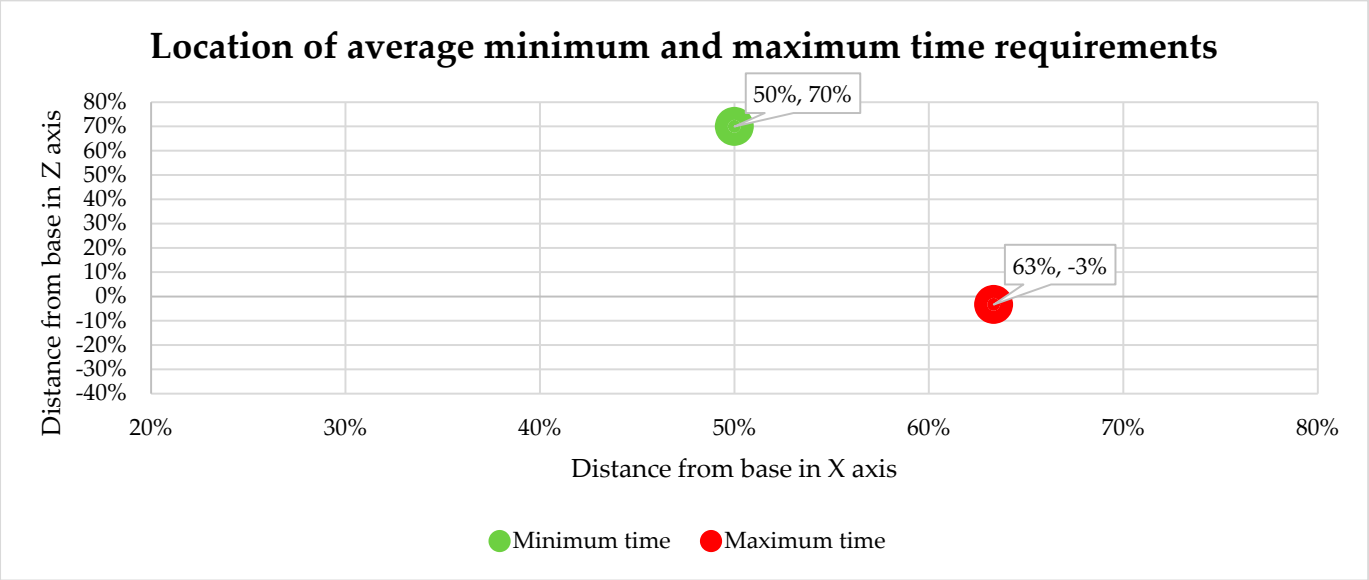


Figure 16. Location of average minimum and maximum time requirements. The chart identifies the average spatial positions for minimum (green marker) and maximum (red marker) time demands in the X and Z axes. These insights support the optimization of robot placement for minimizing operational time in various tasks.

5. Discussion

The experimental results confirmed that positioning the tool center point (TCP) near the middle of the robot's working range leads to lower energy consumption than positions closer to the base or at maximum reach. This finding, which was based on minimizing the moment of inertia, was consistent across robots with similar constructions, suggesting that placing the TCP at approximately 50% of the robot's reach is the most energy-efficient. Due to limited access to multiple ABB industrial robots, all measurements were conducted using the simulation software, with initial verification against data from a physical robot.

An initial sample of seven robots with varying reach ranges from 0.5 to 4.2 m was tested, revealing different optimal positions for minimum energy and time demands. Significant design differences between the robots led to inconsistencies, making the sample unsuitable for the methodology. A refined sample of ABB robots with similar designs and range limits yielded more consistent results, which were suitable for further development.

The study highlights how structural characteristics affect energy consumption, particularly through the moment of inertia, which affects the robot's resistance to rotation. In articulated robotic arms, inertia is affected by the mass distribution in each segment and its distance from the rotation axis. Robots with counterbalancing mechanisms required less energy during vertical movements, whereas robots without such mechanisms showed increased energy demands in extreme vertical positions. This shows that energy efficiency is significantly affected by the design properties, such as counterbalancing mechanisms and mass distribution, especially when frequent gravitational force compensation is required.

A parametrically editable trajectory was created in Siemens Tecnomatix Process Simulate 2402 for consistent testing at various heights and distances, with constant X and Z-axis distances for each point. To increase the energy demand, the trajectory was designed for continuous configuration changes, with constraints limiting axis utilization by 5% to account for real-world component placement inaccuracies.

The experimental verification showed that the principle of minimizing energy consumption around the center of the working range holds across various 6-axis articulated robots, supporting the creation of a general optimization framework for workstations with similar robot designs. A total of 274 trajectories were tested, with 173 meeting the required constraints, suggesting that the ideal X-axis working distance is approximately 50% of the

maximum reach. Adjusting the height along the Z-axis further optimizes the energy consumption: Lowering the Z-axis reduces energy use but increases the time, whereas raising the height decreases the time but raises the energy requirements. Practical applications require a balance between energy and time efficiency based on operational priorities.

These findings provide insights into designing work trajectories that minimize inertia and torque demands, particularly for TCP positioning. Even with a similar trajectory, construction differences significantly influence energy use, highlighting the importance of factoring in structural specifics when optimizing robot positioning.

Although the results suggest that positioning the TCP near the midpoint of the reach optimizes the energy efficiency, differences in the arm configurations and counterbalancing mechanisms affect the energy outcomes. These structural variances should be considered when applying generalized rules.

The two main limitations offer opportunities for further research. First, the study focused on articulated 6-axis robots, primarily an ABB IRB 120. Although these results provide reliable data, future studies could include SCARA or delta robots to broaden their applicability. Second, the simulations were conducted without a load, allowing for a controlled analysis but lacking realistic load-bearing operations. Experiments under a load condition would enhance the estimation of energy consumption.

Based on these findings, a robotic workstation design methodology will be developed, considering the impact of component positioning on energy usage. The theoretical analysis indicates that early-stage layout optimization can reduce energy demands, but current design software lacks energy-based optimization features. Thus, energy considerations have not yet been systematically integrated into robotic workstation design practices.

6. Conclusions

The present paper deals with the reduction in the energy and time costs of selected industrial robots by modifying the relative positions of the robots and their peripherals. It should be emphasized that today's layout of robot station components is based on the main criteria, which are the reachability of all working points and observation of the station's specified working time.

Although a focus is currently placed on reducing energy consumption, a complex algorithm that automates the distribution of components in a robotic workplace to reduce the workplace's power consumption would be beneficial. Then, the theoretical part continues with an overview of the ways in which the energy and time intensity of such workplaces are reduced. This section is divided into three groups, with the first group focusing on software modifications that can be applied only after the station has been completely built.

The most widely used method in science and practice is to change the speed and acceleration of an industrial robot by changing the sequence of individual robotic operations or by optimizing trajectories. The second group deals with hardware modifications of industrial robots and their equipment. This includes modifications such as the matrix selection and design of industrial robots or even power-sharing add-on devices. The third part is the least-investigated part in the field of reducing electric energy consumption. This section deals with the initial stage of the workstation layout by upregulating the relative positions of the components along the X, Y, and Z axes. Although this area has not been under-researched extensively, most research has focused only on reducing working time.

The results confirm that the energy consumption of an industrial robot varies as the position of the robot's basic coordinate system changes with respect to the trajectories. The results show that this position has both positive and negative effects on energy consumption. Based on the evaluation of the experiments, it can be confirmed that the optimal position for the selected sample of robots is the range in the X-axis direction from the robot base, which is 50% of the maximum range of the robot, in a case that the position of the robot base along the Z-axis would be shifted.

Based on simulations and computer models, the optimal working distance for the selected industrial robots in the X-axis was found to be approximately 50% of their maxi-

mum range. This result is valid for both energy and time optimization. Simulations have shown that by maintaining this X-axis distance and manipulating the height on the Z-axis, energy consumption can be influenced. Moving the working position below the level of the robot's base reduces energy consumption but increases the time required to perform the necessary operations.

On the other hand, if the goal is to reduce the time required to perform tasks, the working position must be moved higher above the base, which leads to an increase in energy consumption. From the results of these simulations, a trade-off between time and energy efficiency can be determined, with a specific choice depending on the performance requirements of a given system.

An ABB IRB 120 robot was used to demonstrate the proposed method. The simulation results in ABB RobotStudio 2024.1.1 software demonstrated the effectiveness of the proposed method, and a model of the robot's energy consumption was created based on the identification of the required parameters. The simulated results confirmed that the model achieved satisfactory accuracy. The next phase of research is planned to quantitatively compare the simulated consumption with the measured values to validate the accuracy of the model under real conditions.

The relationship between the robot's position and its energy consumption was also analyzed. The results showed that for optimal energy consumption, the robot position should be carefully balanced. An excessively high or low position has a negative effect on energy efficiency. The trade-off between optimal position and productivity will be the subject of further research.

Further research will examine developing a methodology to incorporate this problem of reducing power consumption based on the relative positioning of industrial robots and their peripherals into the existing methodology for creating robotic workplaces, which also contributes to sustainable manufacturing.

Practical recommendations for further research and applications include, in particular, suggestions for the optimization of robotic workplaces and future research directions. Based on these findings, it is advisable to take into account design differences between robot models, especially balancing mechanisms and arm configurations, when designing workstations. When positioning TCPs, it is recommended to avoid extreme vertical positions that are energy intensive, unless this is impossible.

Further research should focus on a more detailed analysis of the impact of the balancing mechanisms and arm configuration on energy consumption at different trajectories. A detailed study of these design factors can lead to the development of broader rules for optimizing the energy efficiency of various robotic systems.

For vertical robot motions, it is recommended to prefer trajectories that minimize vertical motions, thereby reducing the energy consumption due to gravity loads. For downward motions, energy recuperation might be suitable. For Z-axis motions, it is important to consider changes in the robot's center of gravity, which directly affect the stability and torque requirements, especially for upward motions.

In addition to practical recommendations, this research has the potential to continue in the direction of comparing mathematical models based on theoretical frameworks for robot energy consumption and dynamics. Such an approach could lead to the development of universal optimization models that would be applicable to a wide range of robotic applications, thus providing valuable insights for efficient and energy-saving solutions in industrial automation.

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